

Errata to *Counting Elliptic Curves with a Cyclic Isogeny*

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This separata records the changes made in the corrected edition. Page references are to the physical pages of the submitted thesis PDF, counted from the title page as p. 1; thus thesis p. 1 is Submitted Thesis p. 9. Each entry is a compact source-style diff: a line beginning with `-` is replaced by the corresponding line beginning with `+`. Ellipses suppress unchanged context; enough surrounding text is retained to identify the substitution. Pure insertions use only `+`. Purely new frontmatter announcing the corrected edition is not listed as an erratum.

Submitted Thesis p. 3

```
-... power saving ...
+... power-saving ...
```

Submitted Thesis p. 4

```
-... blessed with support and ...
+... blessed with the support and ...

-... go out to to my ...
+... go out to my ...
```

Submitted Thesis p. 5

```
-... fellow-travelers who ...
+... fellow-travelers whom ...

-... experience has ...
+... experience have ...
```

Submitted Thesis p. 9

```
-... elliptic ...
+... elliptic ...
```

Submitted Thesis p. 13

```
-... \textup{unsigned} ...
+... \textup{an unsigned} ...
```

Submitted Thesis p. 15

```
-... \consteqprm m  $X^{1/6} + O(X^{1/8 + \epsilon})$ 
+... \consteqprm m  $X^{1/6} + O(X^{38/291 + \epsilon})$ 

-... \constadprm m  $X^{1/6} + O(X^{1/8 + \epsilon})$ 
+... \constadprm m  $X^{1/6} + O(X^{38/291 + \epsilon})$ 
```

Submitted Thesis p. 18

```
-$\twerror m$ & $2$ & $6$ & $12$ & $12$ & $12$ & $12$ & $21$ & $24$ & $15$ & $24$ & $36$ & $33$ \\\
+$\twerror m$ & $2$ & $6$ & $12$ & $12$ & $12$ & $12$ & $18$ & $24$ & $51/4$ & $24$ & $36$ & $30$ \\\

-... 7}, \Cref{Theorem: asymptotic for NQ(X) for m = ...
+... 7}, \Cref{Theorem: asymptotic for twN(X) for m = ...

-... and 25}, \Cref{Theorem: asymptotic for NQ(X) for m = 13}, and ...
+... and 25}, \Cref{Theorem: asymptotic for twN(X) for m = 13}, and ...
```

Submitted Thesis p. 20

```
-... m (X)$ associated ...
+... m (s)$ associated ...
```

Submitted Thesis p. 21

-... 5, where our estimates ...
+... 5, where we use our estimates ...

-... and we able to ...
+... and we are able to ...

Submitted Thesis p. 22

-... summarize the contents in the remainder of ...
+... summarize the remainder of ...

Submitted Thesis p. 23

-... $(X)_{\dots}$
+... $(X)_{\dots}$

Submitted Thesis p. 26

-... millenium ...
+... millennium ...

Submitted Thesis p. 27

- $E^{\{u^2\}} : y^2 : x^3 + u^4 \text{ A } \dots$
+ $E^{\{u^2\}} : y^2 = x^3 + u^4 \text{ A } \dots$

Submitted Thesis p. 28

-... = $E^{\{u^2\}} : y^2 : x^3 + u^4 \dots$
+... = $E^{\{u^2\}} : y^2 = x^3 + u^4 \dots$

Submitted Thesis p. 29

-... torsion ([\Cref{Theorem: Mordell's Theorem}](#) and [\Cref{Theorem: Mazur's Theorem}](#)). ...
+... torsion ([\Cref{Theorem: Mazur's Theorem}](#)). ...

-... $(x, -y)_{\dots}$; ...
+... $(x_0, -y_0)_{\dots}$; ...

-... passes though ...
+... passes through ...

Submitted Thesis p. 30

-... `group\plaw{\includegraphics[scale=0.3]{Graphics/y2=x3-3x+62.png}}`{The ...
+... `group\plaw{\includegraphics[scale=0.3]{Graphics/y^2=x^3-3x+62.png}}`{The ...

-... via its the isomorphism with ...
+... via its isomorphism with ...

Submitted Thesis p. 32

-... of $E_{\dots}$, we ...
+... of $E_{1\dots}$, we ...

Submitted Thesis p. 33

-... already gives ...
+... already give ...

-... $u^4 \text{ A } B = \dots$
+... $u^4 \text{ A } B$ and $B = \dots$

-... $E_2 \rightarrow E_2$ with
+... $E_2 \rightarrow E_1$ with

Submitted Thesis p. 35

-... $\subseteq E(\mathbb{Q})$ for ...
+... $\subseteq E_1(\mathbb{Q})$ for ...
-... out that the $\ker \phi$...
+... out that $\ker \phi$...

Submitted Thesis p. 36

- $f^{(c)}(t) = \prod_{\substack{P \in \Phi \\ P \neq \infty}} (t - x(P)) = \prod_{P \in \Phi} (t - c^2 x(P)) \in \mathbb{Q}[t]$
+ $f^{(c)}(t) = \prod_{\substack{P \in \Phi \\ P \neq \infty}} (t - x(P)) = \prod_{\substack{P \in \Phi \\ P \neq \infty}} (t - c^2 x(P)) \in \mathbb{Q}[t]$

Submitted Thesis p. 37

-... and only Φ is ...
+... and only if Φ is ...
-... E or ...
+... E of ...
-... 2-isogeny ...
+... 2 -isogeny ...

Submitted Thesis p. 38

-... order 25, ...
+... order p^2 , ...

Submitted Thesis p. 39

-... be cyclic n -isogenies ...
+... be a cyclic n -isogeny ...
-... There exist (infinitely many) elliptic curves ...
+... There exist elliptic curves ...

Submitted Thesis p. 40

-By \Cref{Proposition: factoring cyclic m-isogenies}, we can understand cyclic m -isogenies as compositions of cyclic ... is precluded by the absence of any isogeny graphs with more than two 5-isogenies.
+ By \Cref{Proposition: factoring cyclic m-isogenies} and \Cref{Theorem: Kenku-Mazer Theorem}, it suffices to consider the following levels. Namely, $m \in \{2,3,4,5,6,8,9,10,12,15,16,18,25,27\}$. The cases $m \in \{2,4,8,16\}$, $m=3$, $m=5$, and $m \in \{6,12\}$ follow by inspection of the graphs T_8 , $L_3(9)$, $L_3(25)$, and SS , respectively \cite[Tables 1,2,4]{Chiloyan-Lozano-Robledo}. For $m=9,27$, the classified graphs contain at most one cyclic subgroup of the relevant order from any vertex. For $m=10,15$, the rectangular graph $R_4(m)$ has unique rational primary components, hence at most one cyclic m -kernel. For $m=18$, the middle vertices of $R_6(18)$ have no cyclic 9 -isogeny, while at a corner the unique 9 - and 2 -primary kernels determine the 18 -kernel. Finally, $L_3(25)$ gives at most one cyclic 25 -kernel \cite[Tables 1,3]{Chiloyan-Lozano-Robledo}.

Submitted Thesis p. 41

-... 12, 16, 18, see ...
+... 12, 16, see ...

Submitted Thesis p. 42

-... 2.428\,650\,6\dots
+... 1.149\,249\,826\dots

Submitted Thesis p. 43

-... for an elliptic curve E (or more generally, of ...
+... of an elliptic curve E (or, more generally, ...

- $\alpha_{\text{hht}_F} \approx 349,068$
+ $\alpha_{\text{hht}_F} \approx 348,716$

Submitted Thesis p. 44

-... power-savings ...
+... power-saving ...

-... the embedding the coordinates ...
+... the embedding of the coordinates ...

-... this paper, when ...
+... this thesis, when ...

Submitted Thesis p. 45

-... this paper, we ...
+... this thesis, we ...

Submitted Thesis p. 46

-... with power-saving asymptotic for ...
+... with power-saving error for ...

Submitted Thesis p. 47

-... cyclic isogenies by ...
+... cyclic isogeny by ...

-... $m(X) \stackrel{\text{Equation: counts}}{\sim}$...
+... $m(X)$ from $\stackrel{\text{Equation: counts}}{\sim}$...

-... that twistNeq ...
+... that NQeq ...

-... that $\text{twistNad } m$...
+... that $\text{NQad } m$...

-The cases $m = \dots$
+The case $m = \dots$

Submitted Thesis p. 48

-... They parameterize ...
+... They parameterized ...

Submitted Thesis p. 49

-... well-behaved functions ...
+... well-behaved function ...

Submitted Thesis p. 50

-... an effective ...
+... an effectively ...

Submitted Thesis p. 51

```

-\begin{theorem}[\cite[Theorem 1.3]{Pizzo-Pomerance-Voight}]\label{Theorem: asymptotics for N(X) = 3}
-   There exist effective computable constants
+\begin{theorem}[\cite[Theorem 1.3, Proposition 2.9]{Pizzo-Pomerance-Voight}]\label{Theorem: asymptotics for N(X) = 3}
+   There exist effectively computable constants

-... 1.742\,501\,704\,06\ldots, ...
+... 1.671\,470\,808\,106\ldots, ...

-... -0.835\,735\,404\,05\ldots
+... -0.871\,250\,852\,030\ldots

```

Submitted Thesis p. 53

```

-... moduler ...
+... modular ...

```

Submitted Thesis p. 55

```

-... = 108$ while ...
+... = 1728$ while ...

-... 1458) ...
+... 5832) ...

```

Submitted Thesis p. 56

```

-y^2=x^3+(A/e^4)x\pm(B/e^3)
+y^2=x^3+(A/e^2)x\pm(B/e^3)

-... only models for ...
+... only twist minimal models in the quadratic twist class of ...

-... \eqref{Example: ...
+... \Cref{Example: ...

```

Submitted Thesis p. 58

```

-... \set{4, 5, 6, 8, 9, 10, ...
+... \set{4, 5, 6, 7, 8, 9, 10, ...

```

Submitted Thesis p. 59

```

-... $\mathbb{Q}$ are precisely ...
+... $\mathbb{Q}$ is precisely the set of ...

```

Submitted Thesis p. 60

```

-... $E$. If $E$ has ...
+... $E$. If $E_{\text{pr}}$ has ...

```

Submitted Thesis p. 61

```

-... equipped with cyclic 4-isogeny ...
+... equipped with a cyclic 4-isogeny ...

-... 1$ \set{t^4 + 268 t^3} ...
+... 1)$ \set{t^4 + 268 t^3} ...

```

Submitted Thesis p. 64

```

-... & $\set{1, 2, \infty}$ ...
+... & $\set{1, 3/2, 2, \infty}$ ...

-... only unique up to ...
+... unique only up to a ...

```

Submitted Thesis p. 65

```
-... 27 B(a, ...
+... 27 \B m(a, ...
```

Submitted Thesis p. 66

```
-... 7(a,b),c^3\B 7(a,b))$.
+... m(a,b),c^3\B m(a,b))$.

-... minimality defect \eqref{Equation: defect powers}. ...
+... minimality defect defined in \eqref{Equation: twist defect powers}. ...
```

Submitted Thesis p. 67

```
-... with a  $m$ -isogeny ...
+... with an  $m$ -isogeny ...

-... parameterized ...
+... parameterized ...

-... 4, 5, 6, 7, 8, 9, 10, 12, 13, 16, 18, 25}.
+... 4, 5, 6, 7, 8, 9, 10, 12, 13, 16, 18, 25}
```

Submitted Thesis p. 68

```
-... cyclic  $n$ -isogenies whose ...
+... cyclic  $m$ -isogenies whose ...
```

Submitted Thesis p. 69

```
-...  $\Phi_2$  in  $\Phi_1$ , ...
+...  $\Phi_2$ , ...

-...  $n > 1$  ...
+...  $n < m$  ...
```

Submitted Thesis p. 70

```
-... and  $Q$  be generators ...
+... and  $Q$  to be generators ...

-...  $aQ$  generates  $\Phi_2$ , ...
+...  $a$  is a unit modulo  $m$ , ...

-... with  $aQ$  ...
+... with  $a^{-1}Q$  ...

-...  $\mathbb{Z} \subseteq \dots$ 
+...  $\mathbb{Z} \not\subseteq \dots$ 

-...  $\mathbb{Z}$ , then the  $\#\text{parent}\{\Phi_1 \dots$ 
+...  $\mathbb{Z}$ , then  $\#\text{parent}\{\Phi_1 \dots$ 
```

Submitted Thesis p. 71

```
- The image of  $\text{Gal}_{\mathbb{Q}}$  in  $\text{GL}_2(\mathbb{Z}/m\mathbb{Z})$  maps each of these vectors to a multiple of itself, and a little linear algebra shows that the set of all such matrices in  $M_2(\mathbb{Z}/m\mathbb{Z})$  is precisely the set
+ The image of  $\text{Gal}_{\mathbb{Q}}$  in  $\text{GL}_2(\mathbb{Z}/m\mathbb{Z})$  preserves each of the cyclic subgroups generated by these two vectors, and hence is contained in the subgroup

- \set{\begin{pmatrix}
- a & b \\ 0 & a + n b + n\text{prm } c
- \end{pmatrix} : a \in (\mathbb{Z}/m\mathbb{Z})^\times \text{ and } b, c \in \mathbb{Z}/m\mathbb{Z}}.
+ H_{\{m,n\}} \colonequals \set{\begin{pmatrix}
+ a & b \\ 0 & d
+ \end{pmatrix} \in \text{GL}_2(\mathbb{Z}/m\mathbb{Z}) : d \equiv a + n b \pmod{n\text{prm}}}.

- Conversely, if the image of  $\text{Gal}_{\mathbb{Q}}$  on  $\text{GL}(E[m](\overline{\mathbb{Q}}))$  is of the form given by \eqref{Equation: Galois image in GLm} for some choice of basis  $P_1, P_2$ , then  $E$  has two unsigned cyclic  $m$ -isogenies with kernels  $\Phi_1$  and  $\Phi_2$  having intersection of size  $n$ .
```

+ Conversely, if the image of $\text{Gal}_{\mathbb{Q}}$ on $E[m](\mathbb{Q})$ is contained in $H_{m,n}$ for some choice of basis P_1, P_2 , then the cyclic subgroups generated by P_1 and P_1+nP_2 are Galois-stable and have intersection of order n . Thus E has two unsigned cyclic m -isogenies with kernels having intersection of size n .

Submitted Thesis p. 72

-... $\mathbb{Q}$, and do not have any $\mathbb{Q}$ -points. ...
+... $\mathbb{Q}$

-... map from image of ...
+... map from the image of ...

-... image of $\text{Gal}_{\mathbb{Q}}$...
+... image of $H_{m,n}$ from $\text{Gal}_{\mathbb{Q}}$...

-... of $(\mathbb{Z}/m\mathbb{Z})^\times$. Thus, for ...
+... of $(\mathbb{Z}/m\mathbb{Z})^\times$. In particular, there are no elliptic curves $E/\mathbb{Q}$ possessing such a pair. Thus, for ...

-... of $\text{Chiloyan-Lozano-Robledo}$. '
+... of $\text{Chiloyan-Lozano-Robledo}$.

-... $\{12, 16, 18\}$. Then ...
+... $\{12, 16\}$. Then ...

Submitted Thesis p. 73

-... the moduli space for the space of ...
+... the moduli space of ...

-... -3 (3 ...
+... $-(3$...

-... n (t) for m , ...
+... n (t) for m , ...

Submitted Thesis p. 76

-... weaker conditions $\mathcal{R}$ than ...
+... weaker conditions on $\mathcal{R}$ than ...

Submitted Thesis p. 77

- $\text{Area}(\text{calR}_m(X)) \neq \text{Area}(\{(X^{1/2 \deg B} a, X^{1/2 \deg B} b) : (a, b) \in \text{calR}_m(1)\})$
+ $\text{Area}(\text{calR}_m(X)) \neq \text{Area}(\{(X^{1/2 \deg B} a, X^{1/2 \deg B} b) : (a, b) \in \text{calR}_m(1)\})$

Submitted Thesis p. 78

-... to $O(X^{1/2 \deg B})$...
+... to $O(X^{1/2 \deg B})$...

Submitted Thesis p. 91

-... -3 , ...
+... $-3b^2$, ...

Submitted Thesis p. 92

-... $\mathcal{L}$, and independent of ...
+... $\mathcal{L}$, and is independent of ...

-... $\sqrt{2} = 4.828\dots$
+... $\sqrt{5} = 6.472\dots$

Submitted Thesis p. 94

```
-... Let  $\sigma, \rho$  in ...
+... Let  $\sigma$  in ...

-...  $t^{-\sigma}$  ...
+...  $u^{-\sigma}$  ...

-... Bingham--Glodie--Teugels ...
+... Bingham--Goldie--Teugels ...
```

Submitted Thesis p. 96

```
-...  $\frac{X}{\zeta(k)}$  ...
+...  $\frac{X}{\zeta(2)}$  ...
```

Submitted Thesis p. 97

```
-... and  $L_{\beta(s)}$  both converge absolutely ...
+... and  $L_{\beta(s)}$  converge absolutely ...
```

Submitted Thesis p. 98

```
-... be complex functions functions of ...
+... be complex functions of ...

- $\mu_{FG}(\sigma) = \dots$ 
+ $\mu_{FG}(\sigma) \leq \dots$ 
```

Submitted Thesis p. 100

```
-... to Roux \cite{Roux}
+... to Roux \cite{Roux}.

-...  $\delta/\xi$  ...
+...  $\delta/\lfloor \xi \rfloor$  ...
```

Submitted Thesis p. 101

```
-... the Euler-Mascheroni constant ...
+... the Euler--Mascheroni constant ...
```

Submitted Thesis p. 102

```
-for any  $\epsilon > 0$ , ...
+For any  $\epsilon > 0$ , ...

-... 0.607927  $X$  ...
+... 0.607927  $y$  ...

-... has a simple pole of ...
+... has a pole of ...

-...  $\mu_{\zeta}(\sigma) = \dots$ 
+...  $\mu_{\zeta}(\sigma) \leq \dots$ 
```

Submitted Thesis p. 104

```
-... for addressing third step ...
+... for addressing the third step ...
```

Submitted Thesis p. 105

```
-... that admits ...
+... that admit ...

-... by  $\text{twistNeq}$  ...
+... by  $\text{twNad}$  ...
```

-... are important difference between ...
+... are important differences between ...

Submitted Thesis p. 106

-... $\{\text{calE}\} (X) = \sum_{n \geq 1} \text{twNad} \dots$
+... $\{\text{calE}\} (X) = \sum_{e \geq 1} \text{twNad} \dots$

-... f), \label{Equation: ...
+... f). \label{Equation: ...

- and applying M^{ob} inversion to $\text{eqref{Equation: cM}(X e)}$ in terms of $\text{twN}(X e)$ yields
- \begin{equation}
- \text{twNad} \{\text{calE}\} (X; e) = \sum_{f \mid X^{1/6}} \mu(f) \text{cM} \text{calE} (e^6 f^6 X; e f).
- \end{equation}
- Note, however, that any $E \in \text{calE}$ with $H(E) > e^6 X$ cannot contribute to $\text{twNad} \{\text{calE}\} (X; e)$, so in fact we have the equality
+
+ For the inversion, we must keep the raw-height cutoff fixed. A model counted by $\text{twNad} \text{calE}(X; e)$ has $\text{tmd}(E) = e$ and $\text{rawheight}(E) \leq e^6 X$. Inclusion--exclusion among the multiples of e therefore gives

Submitted Thesis p. 107

- $\text{twNad} \{\text{calE}\} (X; e) = \sum_{f \mid X^{1/6}} \mu(f) \text{cM} \text{calE} (e^6 X; \dots$
+ $\text{twNad} \{\text{calE}\} (X; e) = \sum_{f \geq 1} \mu(f) \text{cM} \text{calE} (e^6 X; \dots$

-... e f). \label{Equation: ...
+... e f), \label{Equation: ...

-Substituting $\text{eqref{Equation: ...}$
+where the sum is finite. Substituting $\text{eqref{Equation: ...}$

-... a modest ...
+... an absolute-value ...

- For all $e, n \in \mathbb{Z}_{>0}$ and $X > 0$, if $e \mid n$ then we have
- \begin{equation}
- $0 \leq \sum_{e \mid n} \mu(n/e) \text{cM} \text{calE}(e^6 X; n) \leq \text{cM} \text{calE}(n^6 X; n)$.
- \end{equation}
+For all $n \geq 1$ and $X > 0$,
+ \begin{equation}
+ $|\sum_{e \mid n} \mu(n/e) \text{cM} \text{calE}(e^6 X; n)| \leq \sum_{e \mid n} |\mu(n/e)| \text{cM} \text{calE}(e^6 X; n)$.
+ \end{equation}
+Every model on the right has twist height at most X , and after summing over n a model of twist minimality defect D occurs with multiplicity at most $\tau(D)^2$.

-By $\text{eqref{Equation: defining cM}}$, $\text{cM} \text{calE} (X; e)$ counts elliptic curves $E \in \text{calE} \dots > e^6 X$ for all e a proper divisor of n . The claim follows.
+The displayed inequality is the triangle inequality. If E is counted with $e \mid n \mid D$ $\text{colonequals} \text{tmd}(E)$, then $\text{rawheight}(E) \leq e^6 X \leq D^6 X$; its total multiplicity is at most $\sum_{n \mid D} 2^{2\omega(n)} \leq \tau(D)^2$.
+ \end{proof}
+
+ \begin{lemma} \label{Lemma: weighted 2omega divisor factor}
+For fixed $\alpha > 0$, $\beta > 1$, and $\gamma \geq 2$,
+ \begin{equation}
+ $\sum_{n > y} \frac{2^{2\omega(n)}}{\prod_{p \mid n} (1 + p^{-\alpha})} n^{-\beta} = O_{\alpha, \beta} \left(\frac{\log y}{y^{\beta-1}} \right)$.
+ \end{equation}
+ \end{lemma}
+
+ \begin{proof}
+Since $\prod_{p \mid n} (1 + p^{-\alpha}) \leq \sum_{d \mid n} d^{-\alpha}$ and $2^{2\omega(dm)} \leq 2^{2\omega(d)} 2^{2\omega(m)}$, $\text{Cref{$
Corollary: sum of $2^{2\omega(n)}$ gives partial sums $O_{\alpha} (x^{\log x})$; partial summation proves the claim.

Submitted Thesis p. 108

-... elliptic curve in ...
+... elliptic curves in ...

Submitted Thesis p. 109

-... repetition of ...
+... repetition if ...

-... $E \in \text{scrE}$ of twist ...
+... $E \in \text{calE}$ of twist ...

Submitted Thesis p. 111

```
-... $\NQad \calE (s)$.
+... $\NQad \calE (X)$.

-... for $\LQad \calE (s)$. ...
+... for $\NQad \calE (X)$. ...

-... \twistLad \calE (n), \ \hQeq ...
+... \twistLad \calE (s), \ \hQeq ...

-... \hQeq m (n) &\colonequals \hQad \calE, \ \text{and} \ \
+... \hQeq m (n) &\colonequals \hQad \calE (n), \ \text{and} \ \
```

Submitted Thesis p. 112

```
-... $\hQeq ...
+... $\twistheq ...

-... \twistLad \calE (n), \ \hQad ...
+... \twistLad \calE (s), \ \hQad ...

-... \hQad m (n) &\colonequals \hQad \calE, \ \text{and} \ \
+... \hQad m (n) &\colonequals \hQad \calE (n), \ \text{and} \ \

-... number of of elliptic curves ...
+... number of elliptic curves ...
```

Submitted Thesis p. 113

```
-... prove \Cref{Intro Theorem: asymptotic for NQ(X) for 5 < m <= 9} when $m = 7$, with ...
+... prove \Cref{Intro Theorem: asymptotic for twN(X) for m of genus $0$} when $m = 7$, with ...
```

Submitted Thesis p. 115

```
-... discussssed ...
+... discussed ...

-... $7$-groomed, let $\ell$ be prime, and let $v \in \Z_{\geq 0}$. Then the following statements hold.
+... $7$-groomed, let $\ell$ be prime, and let $v \in \Z_{>0}$. Then the following statements hold.
```

Submitted Thesis p. 116

```
-... by \eqref{Equation: defect powers}, ...
+... by \eqref{Equation: twist defect powers}, ...

-so $\ell \nmid \gcd(\tB 7(a,b), \{C 7\}(a, b))$ and thus $\ell^v \mid \tmd(\A 7(a,b), \B 7(a,b))$ if and only if $\ell^{3v} \mid \mid{C 7}(a,b)$. If $\ell = 2$, a short computation confirms that $\B 7(a, b)$ is odd whenever $(a, b)$ is $7$-groomed, so our claim also holds in this case.
+so $\ell \nmid \gcd(\tB 7(a,b), \{C 7\}(a, b))$ and thus $\ell^v \mid \tmd(\A 7(a,b), \B 7(a,b))$ if and only if $\ell^{3v} \mid \mid{C 7}(a,b)$. If $\ell = 2$, then $\C 7(a,b)$ and $\tA 7(a,b)$ are both odd whenever $(a,b)$ is $7$-groomed, so $\A 7(a,b)$ is odd and our claim also holds in this case.

-... 7(b,0) ...
+... 7(a,0) ...

-... $G$ to cancel all occurences of the variable $a$, ...
+... $G$ to cancel all occurences of the variable $a$, ...
```

Submitted Thesis p. 117

```
-... $G$ to cancel all occurences of the variable $b$, ...
+... $G$ to cancel all occurences of the variable $b$, ...
```

Submitted Thesis p. 118

```
-... $(F, G) = \in \set{(\tA ...
+... $(F, G) \in \set{(\tA ...

-... -3\parent{13 ...
+... 3\parent{13 ...
```

```

-... -9\parent{303 ...
+... 9\parent{303 ...

-... 1763 a - ...
+... -1763 a + ...

-          \gcdP {\tB 7} {\C 7} 2 (a, b) &= -\parent{13 a + 20 b}, \ \text{and} \ \
+          \gcdP {\tB 7} {\C 7} 2 (a, b) &= 13 a + 20 b, \ \text{and} \ \

-... a - b), ...
+... a + b), ...

-... \twistdefect(A(a, b), B(a, ...
+... \twistdefect(\A 7(a, b), \B 7(a, ...

```

Submitted Thesis p. 119

```

-... Now it ...
+... Now if ...

-Now let  $p = 2$ . A short computation confirms that for  $a$  and  $b$  coprime,  $\{ \tB 7 \}(a, b) \not\equiv 0 \pmod{2}$  and  $\{ \C 7 \}(a, b) \not\equiv 0 \pmod{2}$ , so our claim holds vacuously in this case.
+Now let  $p = 2$ . For  $a$  and  $b$  coprime,  $\{ \tA 7 \}(a, b) \not\equiv 0 \pmod{2}$  and  $\{ \C 7 \}(a, b) \not\equiv 0 \pmod{2}$ , so  $\{ \A 7 \}(a, b)$  is odd and our claim holds vacuously in this case.

-...  $b$  and cube of ...
+...  $b$  and the cube of ...

```

Submitted Thesis p. 120

```

-... \mid g 7(t) ...
+... \mid f 7(t) ...

```

Submitted Thesis p. 121

```

-\item For all prime  $\ell \neq 3, 7$  ...
+\item For every prime  $\ell \neq 3, 7$  ...

```

Submitted Thesis p. 122

```

-Hensel's lemma still applies to  $\h 7(t)$ : let  $t_0, t_1$  be the roots of  $\h 7(t)$  in  $\Z_7$  with  $t_0 \colonequals 248\,044 \pmod{7^7}$  (so that  $t_1 = -1 - t_0$ ). We claim that
+Hensel's lemma still applies to  $\h 7(t)$ : let  $t_0, t_1$  be the roots of  $\h 7(t)$  in  $\Z_7$  with  $t_0 \colonequals 248\,044 \pmod{7^7}$  (so that  $t_1 = -1 - t_0$ ). The branch  $t_0 \equiv -1 \pmod{7}$  has  $\tg 7(t_0) \not\equiv 0 \pmod{7}$ , while on the branch  $t_1 \equiv 0 \pmod{7}$  one checks that  $7^4 \mid \tf 7(t)$  and  $7^7 \nmid \tg 7(t)$  whenever  $t \equiv 42 \pmod{49}$ . Consequently, for  $v \geq 3$  we have

-\calT 7(7^{3v}) ...
+\calT 7(7^v) ...

-for  $3v \geq 7$ . Indeed,  $\tg 7(t_1) \equiv 0 \pmod{7^7}$ , so we can afford to approximate  $t_1$  modulo  $7^{3v-7}$ . As  $g 7(t_0) \not\equiv 0 \pmod{7}$  and  $g 7(t_1) \not\equiv 0 \pmod{7^8}$ , no other values of  $t$  suffice. Thus  $\calT 7(7^{3v}) = 1 + 7^7 = 823544$ .
+because the first branch requires  $7^{3v} \mid \h 7(t)$ , whereas on the second branch  $7^{3v-7} \mid \h 7(t)$  is necessary and sufficient for the two divisibility conditions defining  $\calT 7(7^v)$ . Thus  $\calT 7(7^v) = 1 + 7^7 = 823544$ .

```

Submitted Thesis p. 123

```

-}\calT {7} (7^v) for  $v \geq 2$ 
+}\calT {7} (7^v) for  $v = 1, 2$ 

-... of  $\bbZ[\zeta_6]$  ...
+... of  $\bbZ[3 \zeta_6]$  ...

```

Submitted Thesis p. 124

```

-...  $\bbZ[\zeta_6]$ . ...
+...  $\bbZ[3 \zeta_6]$ . ...

-...  $\zeta_6$ ,  $e$  ...
+...  $\zeta_6$ ,  $e^k$  ...

```

Submitted Thesis p. 125

```
-... $p$ prime ...  
+... a prime $p$ ...
```

Submitted Thesis p. 127

```
-... defect for m = 7}(c),  
+... defect for m = 7}(b),  
  
-e_0^{12} m^6 ...  
+e_0^{12} n_0^6 ...  
  
-... rounding $m$ ...  
+... rounding $n_0$ ...  
  
-... \cdot 7^7$ ...  
+... \cdot 7^3$ ...
```

Submitted Thesis p. 128

```
-... attained when $t = 0$ ...  
+... attained when $\theta = 0$ ...  
  
-... attained when $t = 4.980\,802\,4\ldots$ ...  
+... attained when $\theta = 4.980\,802\,4\ldots$ ...  
  
-... -0.371\ldots)$$. ...  
+... -0.371\ldots)$$. ...  
  
-... 1))\{C(a, ...  
+... 1))\{C 7(a, ...
```

Submitted Thesis p. 132

```
-... size of order of growth for ...  
+... order of growth of ...  
  
-for $X \geq 2$, $d$, $e \geq 1$.  
+for $X \geq 2$, $e \geq 1$.
```

Submitted Thesis p. 133

```
-... {e^3}, (a, b) ...  
+... {e^3}, (a:b) ...  
  
-... L_7(X; d, ...  
+... L_7(X; e, ...
```

Submitted Thesis p. 134

```
-... 7(x; ...  
+... 7(X; ...  
  
-\cM 7(x;e)  
+\cM 7(X;e)
```

Submitted Thesis p. 135

```
-... 7 (c\prm, d\prm) \ ...  
+... 7 (c\prm, d\prm)) \ ...
```

Submitted Thesis p. 136

```
-... \pm b_0 ...  
+... \pm \abs{b_0} ...  
  
-... b_0)) \leq \parent{\frac{101 + 16 \sqrt{37}}{27}}^{\frac{1}{2}} e_0^{\frac{1}{2}}.  
+... b_0)) \leq e_0^{\frac{1}{2}}.
```

Submitted Thesis p. 137

- These extrema are both attained when $a_0/b_0 = -1/2$.
- + The lower extremum is attained when $a_0/b_0 = -1/2$, and the upper extremum is attained when $b_0 = 0$.

Submitted Thesis p. 140

- ... $\displaystyle\{\frac{2}{p^2 + 1}\}$, ...
- +... $\displaystyle\{\frac{2}{(p+1)^2}\}$, ...
- ... $\frac{2}{p^2 + 1}$
- +... $\frac{2}{(p+1)^2}$

Submitted Thesis p. 142

- By \Cref{Lemma: bound on T(e) for m = 7}(f), $\mathcal{T}_m(e) = O(2^{-\omega(e)})$, so by \Cref{Lemma: asymptotic for M(X; e) for m = 7}, we have
- + By \Cref{Lemma: bound on T(e) for m = 7}(f), \Cref{Lemma: asymptotic for M(X; e) for m = 7}, and \Cref{Proposition: bounding the summands of twN calE X},
- $\mathcal{M}_m(X; e) = O(\text{parent}\{\frac{2^{-\omega(e)}}{X^{1/6}}\}e^3)$.
- + $\text{abs}\{\text{twistNadgy } 7(X)\} \ll X^{1/6} \sum_{n>y} \frac{2^{-\omega(n)}}{n^2} \prod_{p \mid n} (1 + \frac{1}{p})$.
- Now by \Cref{Proposition: bounding the summands of twN calE X}, we see \begin{equation} \text{twistNegy} \dots \text{ of } f/n^\sigma, \text{ we conclude } \begin{equation} \text{twistNegy } m(X) = O(\text{parent}\{\frac{X^{1/6}}{\log y}\}) \end{equation} as desired.
- + The result follows from \Cref{Lemma: weighted 2omega divisor factor} with $\alpha=1$ and $\beta=2$.

Submitted Thesis p. 143

- $\mu_{\text{twistLad } 7}(\sigma) < 13/84$
- + $\mu_{\text{twistLad } 7}(\sigma) \leq \max\{0, 2-12\sigma\}$

Submitted Thesis p. 145

- Finally, we prove part (c). By \Cref{Theorem: absolutely convergent Dirichlet series have mu = 0}, $\mu_{\text{twistLadR } 7}(\sigma) = 0$ for $\sigma > 1/12$, so by \Cref{Proposition: mu acts like a negative valuation} and \Cref{Theorem: muzeta(sigma)},
- + Finally, we prove part (c). For fixed $1/12 < \sigma_0 < 1/6$, \eqref{equation: bound on twLR for m = 7} gives $\mu_{\text{twistLadR } 7}(\sigma_0) \leq 1$, while absolute convergence gives $\mu_{\text{twistLadR } 7}(\sigma) = 0$ for $\sigma > 1/6$. By subconvexity of μ \cite[Theorem II.1.20]{Tenenbaum},
- $\mu_{\text{twistLad } 7}(\sigma) = \mu_{\text{zeta}_6}(\sigma) < 13/84$
- + $\mu_{\text{twistLadR } 7}(\sigma) \leq 2-12\sigma \ll (1/12 < \sigma < 1/6)$.
- for $\sigma > 1/12$. Our claim follows.
- + Since \Cref{Theorem: muzeta(sigma)} gives $\mu_{\text{zeta}_6}(\sigma) \leq \frac{13}{42}(1-6\sigma) < 2-12\sigma$ on the same strip, \eqref{equation: Defining twLR for m = 7} and \Cref{Proposition: mu acts like a negative valuation} yield $\mu_{\text{twistLad } 7}(\sigma) \leq \max\{0, 2-12\sigma\}$ for $\sigma > 1/12$.

Submitted Thesis p. 146

- ... part (a), since $\zeta(s)$ is nonvanishing when $\sigma > 1$, the ratio $\zeta(6s)/\zeta(12s)$ is meromorphic function for $\sigma > 1/12$. But \Cref{Corollary: twL(s) has a meromorphic continuation} ...
- +... part (a), since $\zeta(s)$ is nonvanishing when $\sigma > 1$, the ratio $\zeta(6s)/\zeta(12s)$ is a meromorphic function for $\sigma > 1/12$. But \Cref{Corollary: twL(s) has a meromorphic continuation} ...

Submitted Thesis p. 147

- The sequence $\text{parent}\{\text{hQad } 7(n)\}_{n \geq 1}$ is admissible \textup{(\Cref{Definition: admissible sequences})} with parameters $(1/6, 1/12, 13/42)$.
- + For every δ with $0 < \delta < 7/97$, the sequence $\text{parent}\{\text{hQad } 7(n)\}_{n \geq 1}$ is admissible \textup{(\Cref{Definition: admissible sequences})} with parameters
- + \begin{equation}
- + $\text{parent}\{1/6, \delta, 97\delta/7\}$.
- + \end{equation}
- ... σ_a , so ...
- +... $> 1/6$, so ...

-For (ii), \Cref{Corollary: twL(s) has a meromorphic continuation for m = 7} tells us that $\text{twistLad } 7(s)$ has a meromorphic continuation when $\sigma = \text{repart}(s) > 1/12$; on the other hand, as $\zeta(12s)$ is nonvanishing for $\sigma > 1/12$, we see that $\zeta(6s)/\zeta(12s)$ has a meromorphic continuation to $\sigma > 1/12$, and so (ii) holds with

+For (ii), \Cref{Corollary: twL(s) has a meromorphic continuation for m = 7} tells us that $\text{twistLad } 7(s)$ has a meromorphic continuation when $\sigma = \text{repart}(s) > 1/12$; on the other hand, as $\zeta(12s)$ is nonvanishing for $\sigma > 1/12$, we see that $\zeta(6s)/\zeta(12s)$ has a meromorphic continuation to $\sigma > 1/12$. Since $\delta < 7/97 < 1/12$, condition (ii) holds on the smaller half-plane $\sigma > 1/6 - \delta$. (The only pole of $L_{\text{Qad } 7}(s)/s$ there is the double pole at $s = 1/6$ indicated in \Cref{Theorem: relationship between twistL(s) and L(s) for m = 7}(b).)

+
+For (iii), suppose first that $1/6 - \delta < \sigma \leq 1/6$, and put $d = 1/6 - \sigma$, so $0 \leq d < \delta$. By \Cref{Corollary: twL(s) has a meromorphic continuation for m = 7},

Submitted Thesis p. 148

$-\delta = 1/6 - 1/12 = 1/12$.
+ $\mu_{\text{twistLad } 7}(\sigma) \leq 12d$.

-(The only pole of $L_{\text{Qad } 7}(s)/s$ with $\sigma > 1/12$ is the double pole ... m = 7), $\mu_{\text{twistLad } 7}(\sigma) < 13/84$. Let $\zeta_a(s) = \zeta(as)$. Applying \Cref{Theorem: muzeta(sigma)}, we have
+Writing $\zeta_a(s) = \zeta(as)$, \Cref{Theorem: muzeta(sigma)} gives

$-\mu_{\zeta_6}(\sigma) = \mu_{\zeta_6}(\sigma) < \frac{13}{42} \left(1 - \frac{6}{12}\right) = \frac{13}{84}$ \end{equation} if $\sigma \leq 1/6$, ... the sequence $\left\{ \frac{1}{n} \log L_{\text{Qad } 7}(n) \right\}_{n \geq 1}$ is admissible with final parameter $\xi = 13/42$.
+ $\mu_{\zeta_6}(\sigma) = \mu_{\zeta_6}(\sigma) \leq \frac{13}{42} (1 - 6\sigma) = \frac{13}{7}d$,
+\end{equation}
+and, as $\zeta(12s)^{-1}$ is absolutely convergent for $\sigma > 1/12$, we have $\mu_{\zeta_{12}^{-1}}(\sigma) = 0$. Therefore \Cref{Theorem: relationship between twistL(s) and L(s)}(b) and \Cref{Proposition: mu acts like a negative valuation} imply
+\begin{equation}
+ $\mu_{L_{\text{Qad } 7}}(\sigma) \leq \frac{13}{7}d = \frac{97}{7}d < \frac{97}{7}\delta$.
+\end{equation}
+For $\sigma > 1/6$, all three factors are absolutely convergent, so $\mu_{L_{\text{Qad } 7}}(\sigma) = 0$. Thus condition (iii) holds with $\xi = 97\delta/7$, proving the claim.

Submitted Thesis p. 149

... $X^{1/6} + O(\text{parent}\{X^{1/8 + \epsilon}\})$
+... $X^{1/6} + O(\text{parent}\{X^{38/291 + \epsilon}\})$

- By \Cref{Lemma: Delta NQ(n) is admissible for m = 7}, the sequence $\left\{ \frac{1}{n} \log L_{\text{Qad } 7}(n) \right\}_{n \geq 1}$ is admissible with parameters $\{1/6, 1/12, 13/42\}$. We now apply \Cref{Theorem: Landau's Tauberian theorem} to the Dirichlet series $L_{\text{Qad } 7}(s)$, and our claim follows.
+ Apply \Cref{Theorem: Landau's Tauberian theorem} and let $\delta \rightarrow 7/97$ in \Cref{Lemma: Delta NQ(n) is admissible for m = 7}; the exponent tends to $1/6 - 7/194 = 38/291$.

-We believe that with sufficient care and appropriate hypotheses, the denominator $\lfloor \xi \rfloor + 2$... for $L_{\text{NQad } 7}(X)$, like $\text{twNad } 7(X)$, would be reduced to $O(X^{1/12 + \epsilon})$.
+ The loss from the twist-counting error $O(X^{1/12} \log^5 X)$ to the exponent $38/291$ above comes from the vertical growth of the analytically continued remainder in \eqref{equation: twistLR(s) as an integral for m = 7}. We believe an elementary argument applying Dirichlet's hyperbola method to \Cref{Theorem: relationship between twistL(s) and L(s)}(a) could achieve a comparable, and perhaps modestly improved, error term without this analytic-growth loss.

Submitted Thesis p. 150

... $(X/108)^{1/6}$ (this bound arises ...
+... $\text{parent}\{3^3 \cdot 7^{18} X^{1/6}\}$ (this bound follows ...

... 7(a)). ...
+... 7(b)). ...

... $\gcd(a, b) = 1$, $b > 0$, and $\mathcal{C}_7(a, b)$...
+... $\gcd(a, b) = 1$, and $\mathcal{C}_7(a, b)$...

... $\gcd(a, b) = 1$, $b > 0$, and $\mathcal{C}_7(a, b)$...
+... $\gcd(a, b) = 1$, and $\mathcal{C}_7(a, b)$...

... m), $e_0^{12} m^6 \leq X/108$, ...
+... m) with $4e_0^{12} m^6 \leq 3^3 \cdot 7^{18} X$...

Submitted Thesis p. 151

... $b_e \text{parent}\{-1+3\zeta_6\}^3$...
+... $b_e \text{prm}\text{parent}\{-1+3\zeta_6\}^3$...

... by \ref{Lemma: ...
+... by \Cref{Lemma: ...

-... twice ...
+... twist ...

Submitted Thesis p. 153

-Reorganizing the sum in \Cref{Theorem: relationship} ...
+Reorganizing \Cref{Theorem: relationship} ...

-\NQad 7 (X) = 2 \sum_{n \leq X} \sum_{c \leq (X/n)^{1/6}} \twistad 7 \parent{n/c^6} \abs{\mu(c)}.
+\NQad 7 (X)=2\sum_{n \leq X} \twistad 7 (n) \, \kfree 2 \parent{(X/n)^{1/6}}.

-... in \Cref{Theorem: asymptotic for NQ(X) for m = 7} ...
+... in \Cref{Theorem: asymptotic for twN(X) for m = 7} ...

-... \leq Y \, \, p \equiv 1 \pmod 3 \} \parent{1 + \frac{2}{p^2+1}}
+... \leq Y \, \, p \equiv 1 \pmod 3 \} \parent{1 + \frac{2}{(p+1)^2}}

Submitted Thesis p. 154

-For \$a, b ...
+For \$e, a_0 ...

-... \, p \equiv a_0 ...
+... \, p \leq X, \, p \equiv a_0 ...

-In particular, letting \$Y = 10^{12}\$, we compute
+In particular, letting \$Y = 8 \cdot 10^9\$, we compute

-17.460\,405\,231\,126\,620 < \, Q \, 7 < 17.460\,405\,231\,134\,835
+17.460\,405\,236\,640\,390 < \, Q \, 7 < 17.460\,405\,236\,985\,885.

-This computation took approximately \$9\$ CPU days. % 789737 seconds.

Submitted Thesis p. 155

-We find \$r_7 \colonequals 243\,228\,665\,965\$ of our first \$s_7 \colonequals 595\,055\,000\,000\$ samples lie in \$\mathcal{R} \, 7\$, so
-\begin{equation}
-\mathcal{R} \, 7 \approx 0.105612 \cdot \frac{r_7}{s_7} = 0.04316889 \ldots
-\end{equation}
-with standard error
+We find \$r_7 \colonequals 243\,228\,665\,965\$ of our first \$s_7 \colonequals 595\,055\,000\,000\$ samples lie in \$\mathcal{R} \, 7\$, so

-0.105612 \cdot \sqrt{\frac{r_7(s_7 - r_7)}{s_7^3}} < 6.8 \cdot 10^{-8}.
+\mathcal{R} \, 7 \approx 0.105612 \cdot \frac{r_7}{s_7} = 0.043168893413 \ldots

-This took 11 CPU weeks to compute. Thus \$\twconst 7 = 0.45822276 \ldots\$, with error bounded by \$6.6 \cdot 10^{-7}\$\$.
+with standard error \$6.73053 \ldots \cdot 10^{-8}\$. This took 11 CPU weeks to compute. Using the midpoint of our interval for \$\mathcal{Q} \, 7\$,
we obtain \$\twconst 7 \approx 0.4582228478 \ldots\$ with Monte Carlo standard error \$7.15 \cdot 10^{-7}\$\$.

-We have the identity \$\constad 7 = \twconstad 7 / 3 \zeta(2)\$, so \$\constad 7 = 0.092\,855\,36 \ldots\$ with an error of \$6.02 \cdot 10^{-8}\$
+We have the identity \$\constad 7 = \twconstad 7 / 3 \zeta(2)\$, so \$\constad 7 \approx 0.092\,855\,36263 \ldots\$ with Monte Carlo
standard error \$1.45 \cdot 10^{-7}\$\$.

-We now turn our attention to \$\constadpm 7\$. As an intermediate step, we wish ... that for some \$M > 0\$ and for all \$u > X\$, we
have
+We now turn our attention to \$\constadpm 7\$. Truncating \eqref{Equation: Definition of ell0 for m = 7} at \$X=10^{42}\$ and using our
list of \$4\,582\,079\$ twist-minimal curves gives

-\abs{\twistNad 7(u) - \twconst 7 \floor{u^{1/6}}} < M u^{1/12} \log^5 u.
+\ell_{7,0} \approx -0.463\,534\,394.

Submitted Thesis p. 156

-Thus
+In \Cref{Theorem: asymptotic for twN(X) for m = 7}, we have shown that for some \$M>0\$ and all \$u>X\$,
+\begin{equation}
+\abs{\twistNad 7(u) - \twconst 7 \floor{u^{1/6}}} < M u^{1/12} \log^5 u.
+\end{equation}
+Thus the omitted tail in \$\ell_{7,0}\$ satisfies

-&\abs{\int_X^\infty \parent{\twistNad 7(u) - \twconst 7 \floor{u^{1/6}}} u^{-7/6} \, \mathrm{d}u} \, \, \\-&\qquad \qquad < M \int_X^\infty u^{-13/12} \log^5 u \, \mathrm{d}u \, \, \\-&\qquad \qquad = 12 M X^{-1/12} (\log^5 X + 60 \log^4 X + 2880 \log^3 X \, \, \\-&\qquad \qquad + 103680 \log^2 X + 2488320 \log X + 29859840);
+&\abs{\frac{16}{\int_X^\infty \parent{\twistNad 7(u) - \twconst 7 \floor{u^{1/6}}} u^{-7/6} \, \mathrm{d}u} \, \,

+&\quad<2M $\{-1/12\}$ (\log⁵X+60\log⁴X+2880\log³X\

+&\quad\quad+103680\log²X+2488320\log X+29859840).

-this gives us a bound on our truncation error.

-We do not know the exact value for M , but empirically, we find that for $1 \leq u \leq 10^{42}$, we have

+We do not know the exact value of M . For the large values of u examined in our computation up to 10^{42} , we find

--5.11 $\cdot 10^{-6} \leq \frac{\text{twistNad } 7(u) - \text{twconst } 7 \lfloor u^{1/6} \rfloor}{u^{1/12} \log^5 u} \leq 6.29 \cdot 10^{-7}$.

+5.11 $\cdot 10^{-6} \leq \frac{\text{twistNad } 7(u) - \text{twconst } 7 \lfloor u^{1/6} \rfloor}{u^{1/12} \log^5 u} \leq 6.29 \cdot 10^{-7}$.

-If we assume these bounds continue to hold for larger u , we find the truncation error for $\ell_7(0)$ is bounded by 68.95 , which catastrophically dwarfs our initial estimate.

+If we assume these inequalities continue to hold for $u > 10^{42}$, the resulting tail error is less than 56.73 .

-We can do better by sidestepping the logarithms. We know that $\text{twistNad } 7(X) - \text{twconst } 7 X^{1/6} = O(X^{1/12 + \epsilon})$ for every $\epsilon > 0$. We let $\epsilon = 10^{-4}$, and find that for $1 \leq u \leq 10^{42}$,

+We can do better by sidestepping the logarithms. Taking $\epsilon = 10^{-4}$, over the values examined up to 10^{42} we find

--1.2174 $\leq \frac{\text{twistNad } 7(u) - \text{twconst } 7 \lfloor u^{1/6} \rfloor}{u^{1/12 + \epsilon}} \leq 0.52272$.

+1.2174 $\leq \frac{\text{twistNad } 7(u) - \text{twconst } 7 \lfloor u^{1/6} \rfloor}{u^{1/12 + \epsilon}} \leq 0.52272$.

-If we assume these bounds continue to hold for larger u , we get an estimated truncation error of $2.43 \cdot 10^{-5}$, which is much more manageable.

+If we assume these inequalities continue to hold for larger u , the resulting tail error is less than $7.79 \cdot 10^{-4}$. Propagating the Monte Carlo standard error in $\text{twconstad } 7$ through the truncated integral contributes $1.09 \cdot 10^{-5}$, giving reported uncertainties 56.73 or $7.90 \cdot 10^{-4}$, respectively.

-Our estimate of $\ell_7(0)$ is also skewed by our estimates of $\text{twconstad } 7$. An error of ϵ in our estimate for $\text{twconstad } 7$ induces an error of

+Using [Equation: constants for m = 7](#), we obtain

- $\frac{\epsilon}{\log X} \int_1^X \frac{\lfloor u^{1/6} \rfloor}{u^{7/6}} du < \frac{\epsilon}{\log X} \int_1^X \frac{u^{-1}}{u} du = \frac{\epsilon}{\log X}$... asymptotic for $NQ(X)$ for $m = 7$. As a sanity check, we verify that

+ $\text{constadpr } 7 \approx -0.164, 049, 781$,

+ $\text{end{equation}}$

+with reported uncertainties 68.98 or $9.59 \cdot 10^{-4}$ under the two extrapolations above. As a sanity check,

Submitted Thesis p. 157

- $\frac{NQ(10^{42})}{10^7} - 42 \text{constad } 7 \log 10 = -0.164, 186, 667 \dots \approx \text{constadpr } 7$,

+ $\frac{NQ(10^{42})}{10^7} - 42 \text{constad } 7 \log 10 = -0.164, 192, 299 \dots \approx \text{constadpr } 7$,

Submitted Thesis p. 158

... Theorem: asymptotic for $twN(X)$ for m of nonzero genus ([Theorem: asymptotic for \$twN\(X\)\$ for \$m = \dots\$](#))

+... Theorem: asymptotic for $twN(X)$ for m of genus 0 ([Theorem: asymptotic for \$twN\(X\)\$ for \$m = \dots\$](#))

... forces us change how ...

+... forces us to change how ...

... (t). In [Section: Establishing notation when \$m = 10\$ and ...](#)

+... (t). In [Section: Estimating \$twN\(X\)\$ for \$m = 5\$ and \$10\$ and ...](#)

... to prove [Intro Theorem: asymptotic for \$NQ\(X\)\$ for \$m > 9\$](#) for $m = 10, 25$

+... to prove [Intro Theorem: asymptotic for \$twN\(X\)\$ for \$m\$ of genus \$0\$](#) for $m = 10, 25$

Submitted Thesis p. 160

... $(a/b, \dots)$

+... $(a/b, \dots)$

Submitted Thesis p. 161

... m -groomed, let ℓ be prime, and let $v \in \mathbb{Z}_{\geq 0}$. Then the following statements hold.

+... m -groomed, let ℓ be prime, and let $v \in \mathbb{Z}_{>0}$. Then the following statements hold.

... implies $\ell \mid$...

+... implies $\ell^v \mid$...

... $\{25\}(t), \{10\}(t)$...

+... $\{25\}(t), \{25\}(t)$...

Submitted Thesis p. 162

```
-... m (t ...
+... m (t) ...

-... / e^2 \bbZ)^2$! ...
+... / e^3 \bbZ)^2$! ...
```

Submitted Thesis p. 163

```
-... \tmd(A(a, b), B(a, ...
+... \tmd(\A m(a, b), \B m(a, ...
```

Submitted Thesis p. 164

```
-... of \ref{Lemma: ...
+... of \Cref{Lemma: ...

-... \tg {10} ...
+... \tg {m} ...

-... \tf {10} ...
+... \tf {m} ...

-\tf {25} (t_1) \equiv \tg {25} (t_1) \equiv 0 \pmod {5^9} \ \ \text{but} \ \ \tg {25} (t_1) \not\equiv 0 \pmod {5^9}.
+\tf {25} (t_1) \equiv \tg {25} (t_1) \equiv 0 \pmod {5^9} \ \ \text{but} \ \ \tf {25} (t_1) \not\equiv 0 \pmod {5^{10}}.
```

Submitted Thesis p. 165

```
-Finally, part (f). From (d)--(e) we conclude that for  $m = 5, 10$ , we have
+Finally, part (f). From (d)--(e), for  $m = 5, 10$  the local factor at  $5^5$  is at most  $3126$ , the local factor at  $2^5$  is at most  $2^5$ ,
, and every other local factor is at most  $2^5$ . Thus

-\cT {m}(e) \leq 3126 \cdot \prod_{\substack{\ell \mid e \\ \ell \neq 5}} \left(1 + \frac{1}{\ell}\right) \leq 1563 \cdot 2^{\omega(e)},
-\end{equation}
-and for  $m = 25$  we have
+\cT {m}(e) \leq 2 \cdot 3126 \cdot 2^{\omega(e)}.
+\end{equation}
+Similarly, for  $m = 25$ , the local factors at  $2^5$  and  $5^5$  are at most  $8$  and  $1953126$ , respectively, so

-\cT {25}(e) \leq 8 \cdot 1953126 \cdot \prod_{\substack{\ell \mid e \\ \ell \neq 2, 5}} \left(1 + \frac{1}{\ell}\right) \leq
7812504 \cdot 2^{\omega(e)},
+\cT {25}(e) \leq 8 \cdot 1953126 \cdot 2^{\omega(e)}.

-%If  $3^5$  or  $7^5$  divides  $e$ , we at worst introduce a factor of  $(27/2)(823544/2)=5558922$ , so in the general case, we see
-%\begin{equation}
-%T(e) \leq \frac{27}{2} \cdot 823544 \cdot 2^{\omega(e)} = 5558922 \cdot 2^{\omega(e)} = O(2^{\omega(e)})
-%\end{equation}
-so  $\cT m(e) = O(2^{\omega(e)})$  as claimed.
+Hence  $\cT m(e) = O(2^{\omega(e)})$  as claimed.
```

Submitted Thesis p. 167

```
- \repr 5 = \repr {10} : (a, b) \mapsto \begin{pmatrix}
- a & 2b \\
+ \repr {25} : (a, b) \mapsto \begin{pmatrix}
+ a & b \end{pmatrix}
```

Submitted Thesis p. 170

```
-\&225000000 a^{16} - 768750000 a^{14} + 1004103125 a^{12} \ \ \&- 601050000 a^{10} + 139912500 \dots \&- 2 \cdot 3 \cdot 89 \cdot 8429 \cdot
\cdot b^2 + 5^2 \cdot 17^2
+ \&253125 a^{12} - 478125 a^{10} + 480625 a^8 - 455000 a^6 \ \
+ \&+ 252200 a^4 - 49360 a^2 + 16 \ \
+ =\& 3^4 \cdot 5^5 \cdot a^{12} - 3^2 \cdot 5^5 \cdot 17 \cdot a^{10} + 5^4 \cdot 769 \cdot a^8 \ \
+ \&- 2^3 \cdot 5^4 \cdot 7 \cdot 13 \cdot a^6 + 2^3 \cdot 5^2 \cdot 13 \cdot 97 \cdot a^4 - 2^4 \cdot 5
\cdot 617 \cdot a^2 + 2^4, \ \ \text{and} \ \
+ \&253125 b^{12} - 1040625 b^{10} + 1886875 b^8 - 1748750 b^6 \ \
+ \&+ 786575 b^4 - 140665 b^2 + 3481 \ \
+ =\& 3^4 \cdot 5^5 \cdot b^{12} - 3^2 \cdot 5^5 \cdot 37 \cdot b^{10} + 5^4 \cdot 3019 \cdot b^8 \ \
+ \&- 2 \cdot 5^4 \cdot 1399 \cdot b^6 + 5^2 \cdot 73 \cdot 431 \cdot b^4 - 5 \cdot 7 \cdot 4019 \cdot b^2 +
59^2
```

Submitted Thesis p. 172

-... 10, 25}\$, the arguments ...
+... 10, 25}\$, consider the arguments ...

- $\text{rawheight}(\text{A}_{\{m\}}(a, b), \text{B}_{\{m\}}(a, b)) / \text{C}_{\{10\}}(a, b)^6$
+ $\text{rawheight}(\text{A}_{\{m\}}(a, b), \text{B}_{\{m\}}(a, b)) / \text{C}_{\{m\}}(a, b)^{\deg B_{\{m\}}}$.

- have $27 |\text{B}_{\{m\}}(a, b)|^2 > 4 |\text{A}_{\{m\}}(a, b)|^3$. For the reader's information,
+ At these arguments, we have $27 |\text{B}_{\{5\}}(a, b)|^2 > 4 |\text{A}_{\{5\}}(a, b)|^3$ when $m = 5$, $4 |\text{A}_{\{10\}}(a, b)|^3 > 27 |\text{B}_{\{10\}}(a, b)|^2$ when $m = 10$, and $4 |\text{A}_{\{25\}}(a, b)|^3 = 27 |\text{B}_{\{25\}}(a, b)|^2$ when $m = 25$. For the reader's information,

Submitted Thesis p. 173

-... be a m -groomed ...
+... be an m -groomed ...

-where $5^6 / 2^{\frac{1}{3}} \cdot 3^{\frac{1}{2}} = 7\,160.050\dots$, and
+where $5^6 / (2^{\frac{1}{3}} \cdot 3^{\frac{1}{2}}) = 7\,160.050\dots$, and

-... $\frac{2^{\frac{211}{90}} \cdot 5^6}{3^{\frac{1}{12}}}$...
+... $\frac{2^{\frac{7}{3}} \cdot 5^6}{3^{\frac{1}{10}}}$...

-where $2^{\frac{211}{90}} \cdot 5^6 / 3^{\frac{1}{12}} = 72\,411.579\dots$.
+where $2^{\frac{7}{3}} \cdot 5^6 / 3^{\frac{1}{10}} = 70\,552.307\dots$.

Submitted Thesis p. 174

-... $25(c)$,
+... $25(b)$,

-... admitting a m -isogeny ...
+... admitting an m -isogeny ...

Submitted Thesis p. 175

-... $X^{\frac{1}{18}}$.
+... $X^{\frac{1}{30}}$.

Submitted Thesis p. 176

-... $(X; e) = \sum_{\substack{d \mid 11 \\ X^{\frac{1}{2} \deg B_{\{m\}}} \mid \gcd(d, e)}} \dots$
+... $(X; e) = \sum_{\substack{d \mid 11 \\ X^{\frac{1}{2} \deg B_{\{m\}}} \mid \gcd(d, e)}} \dots$

Submitted Thesis p. 177

-... $X^{\frac{1}{2 \deg B_{\{m\}}}} \{d \dots$
+... $X^{\frac{1}{2 \deg B_{\{m\}}}} \{d \dots$

-... $X, d, e \geq 1$.
+... $X, d, e \geq 1$,

-... $0 \left(\frac{2^{\omega(e)}}{X^{\frac{1}{2 \deg B_{\{m\}}}} \right) \dots$
+... $0 \left(\frac{2^{\omega(e)}}{X^{\frac{1}{2 \deg B_{\{m\}}}} \right) \dots$

- for $X \geq 2$, $d, e \geq 1$.
+ for $X \geq 2$ and $e \geq 1$.

-... partial proof. ...
+... partial proofs. ...

Submitted Thesis p. 178

-... $\{e^2\}, (a, b) \dots$
+... $\{e^2\}, (a:b) \dots$

-... $\left(\frac{X^{\frac{1}{2 \deg B_{\{m\}}}}}{e^2} \right) \dots$
+... $\left(\frac{X^{\frac{1}{2 \deg B_{\{m\}}}}}{e^2} \right) \dots$

-... $0 \left(\frac{X^{\frac{1}{2 \deg B_{\{m\}}}}}{e^2} \right) \dots$

```
+... 0\parent{\frac{X^{1/(2 \deg B m)}}{e^2}} ...

-... X^{1/2 \deg B m},
+... X^{1/(2 \deg B m)},

-... X^{1/2 \deg B m}\{d\}.
+... X^{1/(2 \deg B m)}\{d\}.
```

Submitted Thesis p. 179

```
-... m(x; e) = & \sum_{\substack{d \mid 11 X^{1/2 \deg B m} \\ \gcd(d, e) = 1}} X^{1/(2 \deg B m)} ...
+... m(X; e) = & \sum_{\substack{d \mid 11 X^{1/(2 \deg B m)} \\ \gcd(d, e) = 1}} X^{1/(2 \deg B m)} ...

-... \frac{1}{\ell} \sum_{\substack{d \mid 11 X^{1/2 \deg B m} \\ \gcd(d, e) = 1}} X^{1/(2 \deg B m)} ...
+... \frac{1}{\ell} \sum_{\substack{d \mid 11 X^{1/(2 \deg B m)} \\ \gcd(d, e) = 1}} X^{1/(2 \deg B m)} ...

-... \frac{X^{1/2 \deg B m}}{d} \sum_{\substack{d \mid 11 X^{1/2 \deg B m} \\ \gcd(d, e) = 1}} X^{1/(2 \deg B m)} ...
+... \frac{X^{1/(2 \deg B m)}}{d} \sum_{\substack{d \mid 11 X^{1/(2 \deg B m)} \\ \gcd(d, e) = 1}} X^{1/(2 \deg B m)} ...

-... 0(X^{-1/2 \deg B m})
+... 0(X^{-1/(2 \deg B m)})

-... X^{1/2 \deg B m}
+... X^{1/(2 \deg B m)}

-... m(x; e) ...
+... m(X; e) ...

-... 0(2^{\omega(e)} X^{1/2 \deg B m}) ...
+... 0(2^{\omega(e)} X^{1/(2 \deg B m)}) ...
```

Submitted Thesis p. 181

```
- Let $m \in 10, 25$. Suppose $y \mid 11 X^{1/2 \deg B m}$. Then
+ Let $m \in \{10, 25\}$. Suppose $2 \leq y \leq X$. Then

-... m \log y \{y^{1-6/\deg B m}, X^{1/2 \deg B m} y^{3/\deg B m} \log X \log^2 y \}
+... m \log y \{y^{1-6/\deg B m}, X^{1/(2 \deg B m)} y^{3/\deg B m} \log X \log^2 y \}

-... e^{3/\deg B m} X^{1/2 \deg B m} \log X \{n\}.
+... e^{3/\deg B m} X^{1/(2 \deg B m)} \log X \{n\}.

-... e^{3/\deg B m} X^{1/2 \deg B m} \log X \{n\}.
+... e^{3/\deg B m} X^{1/(2 \deg B m)} \log X \{n\}.
```

Submitted Thesis p. 182

```
-... \frac{1}{\ell} = 0\parent{\frac{2^{\omega(n)}}{n^{2-6/\deg B m}}} = 0\parent{\sum_{n > ...} \frac{1}{\ell}} = 0\parent{\sum_{n > ...} ...}

-... &= \frac{R^7 X^{1/6}}{m X^{1/\deg B m} \zeta(2)} \parent{Q \{m\}} ...
+... &= \frac{R^m X^{1/\deg B m}}{m X^{1/\deg B m} \zeta(2)} \parent{Q \{m\}} ...

-... X^{1/6} + 0\parent{\frac{X^{1/6} \log y}{y^{1-\deg B m/6}}}
+... X^{1/\deg B m} + 0\parent{\frac{X^{1/\deg B m} \log y}{y^{1-6/\deg B m}}}

-As the summands of \eqref{equation: sum for Q for m = 7} constitute a nonnegative multiplicative arithmetic function, we can factor
  $Q \{m\}$ ...
+As the summands of \eqref{equation: sum for Q for m = 10 and 25} constitute a nonnegative multiplicative arithmetic function, we can
  factor $Q \{m\}$ ...
```

Submitted Thesis p. 183

```
-... &\colonequals \sum_{a \geq 0} \frac{\varphi_6(\deg B a)}{a} ...
+... &\colonequals 1 + \sum_{a \geq 1} \frac{\varphi_6(\deg B a)}{a} ...

-... \left(2+2^{2/3}\right), & \text{if } m = 25 \text{ and } p=5; \}
```

```

+... \left(2+2^{2/3}\right), & \text{if $m = 25$ and $p=2$;} \\
-... {10}(2^2)}{\parent{1 ...
+... {25}(2^2)}{\parent{1 ...

-... \ll X^{1/2} \degB m}$, ...
+... \leq X$, ...

-
      &\sum_{n \leq y} \sum_{e \mid n} \mu \parent{n/e} \parent{\frac{2^{f(\omega(n))}}{e^{3/\degB m}}} X^{1/2 \degB m} \log X \{n
-
      = &\parent{X^{1/2 \degB m} \log X \sum_{e \leq y} \frac{2^{f(\omega(e))}}{e^{1-3/\degB m}}} \sum_{f \leq y/e} \frac{2^{f(\omega(f))}}{f}\}.
+
      &\sum_{n \leq y} \sum_{e \mid n} \mu \parent{n/e} \parent{\frac{2^{f(\omega(n))}}{e^{3/\degB m}}} X^{1/(2 \degB m)} \log X
-
      = &\parent{X^{1/(2 \degB m)} \log X \sum_{e \leq y} \frac{2^{f(\omega(e))}}{e^{1-3/\degB m}}} \sum_{f \leq y/e} \frac{2^{f(\omega(f))}}{f}\}.

```

```

%0\parent{X^{1/2 \degB m}} ...
+%0\parent{X^{1/(2 \degB m)}} ...

... \frac{2^{\omega(f)}}{f} \ \ = 0\parent{X^{1/2 \degB m}} \log X ...
+... \frac{2^{\omega(f)}}{f} \ \ = 0\parent{X^{1/(2 \degB m)}} \log X ...

... = 0\parent{X^{1/2 \degB m}} y^{3/\degB} ...
+... = 0\parent{X^{1/(2 \degB m)}} y^{3/\degB} ...

... p^{6/\degB m}} \}, \ \ \%sum_n \geq ...
+... p^{6/\degB m}} \}, \%sum_n \geq ...

```

```

- Now by \Cref{Proposition: bounding the summands of twN calE X}, we see
+ Set  $\alpha=6/\deg B$ . By \Cref{Proposition: bounding the summands of twN calE X},

- \twistNadgy m(X) =  $O(\sum_{n > y} \frac{2^{-\omega(n)} X^{1/\deg B m}}{n^{2-6/\deg B m}})$ .
+\begin{aligned}
+ & \abs{\twistNadgy m(X)} \\
+ & \leq \sum_{n > y} \sum_{e \mid n} \abs{\mu(n/e)} \\
+ & \frac{2^{\omega(n)} (e^6 X)^{1/\deg B m}}{n^2} \\
+ & = X^{1/\deg B m} \sum_{n > y} \\
+ & \frac{2^{\omega(n)}}{n^{2-\alpha}} \\
+ & \prod_{p \mid n} (1+p^{-\alpha}).
+\end{aligned}

- Combining \Cref{Corollary: sum of 2^omega(n)} and \Cref{Corollary: tail of sum of f/n^sigma}, we conclude
+ Since  $m=10, 25$  gives  $0 < \alpha < 1$ , \Cref{Lemma: weighted 2omega divisor factor} with  $\beta=2-\alpha > 1$  yields

- \twistNad m(X) =  $\twconst m X^{1/\deg B m} + O(\frac{X^{1/2(\deg B m - 3)} \log^{(\deg B m + 3) / (\deg B m - 3)} X}{(\deg B m - 3)})$ 
+ \twistNad m(X) =  $\twconst m X^{1/\deg B m} + O(\frac{X^{1/(2(\deg B m - 3))} \log^{3(\deg B m - 5)/(\deg B m - 3)} X}{(\deg B m - 3)})$ 

- Let  $m = 10, 25$ , and let  $\epsilon$  be a positive quantity with  $\epsilon \ll X^{1/2 \deg B m}$ ; in particular,  $\log y \ll \log X$ . \Cref{
Lemma: asymptotic for twN<=y(X)} for  $m = 10$  and  $25$  and \Cref{Lemma: bound on twN>y(X)} for  $m = 10$  and  $25$  together tell us
+ Let  $m = 10, 25$ . \Cref{Lemma: asymptotic for twN<=y(X)} for  $m = 10$  and  $25$  and \Cref{Lemma: bound on twN>y(X)} for  $m = 10$  and
25 together tell us

```

```

-\twistNad 7(X) = \twconst m ...
+\twistNad m(X) = \twconst m ...

... m} \log y\{y^{\{1 - 6/\deg B m\}}, X^{\{1/2 \deg B m\}} y^{\{3/\deg B m\}} \log X \log^2 y \}.
+... m} \log y\{y^{\{1 - 6/\deg B m\}}, X^{\{1/(2 \deg B m)\}} y^{\{3/\deg B m\}} \log X \log^2 y \}.

-      We let $y = X^{\{1/2(\deg B m - 3)\}} \log^{\{2 \deg B m/(\deg B m - 3)\}} X$, so
+      We let

-      \frac{X^{\{1/\deg B m\}} \log y\{y^{\{1 - 6/\deg B m\}} \asymp X^{\{1/2 \deg B m\}} y^{\{3/\deg B m\}} \log X \log^2 y \asymp X^{\{1/2(\deg B m - 3)\}} \log^{\{(\deg B m + 3) / (\deg B m - 3)\}} X,
+      y = X^{\{1/(2(\deg B m - 3))\}} \log^{\{-2\deg B m/(\deg B m - 3)\}} X.

-      and we conclude
+      For sufficiently large $X$, this choice satisfies $2 \leq y \leq X$. Moreover, $\log y \asymp \log X$, and a direct
      calculation gives

-      \twistNad m(X) = \twconst m X^{\{1/\deg B m\}} + O\parent{X^{\{1/2(\deg B m - 3)\}} \log^{\{(\deg B m + 3) / (\deg B m - 3)\}} X}

```

```

+      \frac{X^{1/\deg B m} \log y}{y^{1 - 6/\deg B m}} \asymp X^{1/(2 \deg B m)} y^{3/\deg B m} \log X \log^2 y \asymp X^{1/(2(\deg B m - 3))} \log^3(\deg B m - 5)/(\deg B m - 3) X.

-      as desired.
+      The finitely many smaller values of  $\beta$  may be absorbed into the implicit constant, proving the claim.

```

Submitted Thesis p. 187

```

-\subsection*{$L$-series}\label{Subection: L-series for m = 10 and 25}\label{Subsection: L-series for m = 10 and 25}
+\subsection*{$L$-series}\label{Subsection: L-series for m = 10 and 25}

-      Let  $m = 10, 25$ . The following statements hold.
+      Let  $m = 10, 25$ , and put
+      \begin{equation}
+      \beta_m \colonequals \frac{1}{2(\deg B m - 3)}.
+      \end{equation}
+      The following statements hold.

-... 1/2(\deg B m - 3)}. ...
+... \beta_m}. ...

```

Submitted Thesis p. 188

```

-      \item We have
-\begin{equation}
-      \mu_{\text{twistLad } m}(\sigma) < 13/84
-      \end{equation}
-      for  $\sigma > 1/(2 \deg B m - 3)$ .
+      \item For  $\sigma > \beta_m$ , we have
+      \begin{equation}
+      \mu_{\text{twistLad } m}(\sigma) \leq \max\left\{0, \frac{1}{\deg B m - \sigma} - \frac{1}{\deg B m - \beta_m}\right\}.
+      \end{equation}

-      The proof is structurally identical to the one given for \Cref{Corollary: twL(s) has a meromorphic continuation for m = 7}.
+      The proof is structurally identical to the one given for \Cref{Corollary: twL(s) has a meromorphic continuation for m = 7},
      with  $1/6$  and  $1/12$  replaced by  $1/\deg B m$  and  $\beta_m$ .

-... \ref{Section: ...
+... \Cref{Section: ...

```

Submitted Thesis p. 189

```

-... For (a), since  $\zeta(s)$  is nonvanishing when  $\sigma > 1$ , the ratio  $\zeta(6s)/\zeta(12s)$  is meromorphic function for  $\sigma > 1/12$ . But \Cref{Corollary: twL(s) has a meromorphic continuation ...
+... For (a), since  $\zeta(s)$  is nonvanishing when  $\sigma > 1$ , the ratio  $\zeta(6s)/\zeta(12s)$  is a meromorphic function for  $\sigma > 1/12$ . But \Cref{Corollary: twL(s) has a meromorphic continuation ...

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Submitted Thesis p. 190

```

-we indeed have  $\mathcal{L}_{\text{Qad } 7}(n) \in \mathbb{Z}_{\geq 0}$ .
+we indeed have  $\mathcal{L}_{\text{Qad } m}(n) \in \mathbb{Z}_{\geq 0}$ .

-      and by \Cref{Theorem: product of Dirichlet series converges} this series converges absolutely for  $\sigma > 1/6$ , so the
      abscissa of absolute convergence for  $\mathcal{L}_{\text{Qad } m}(s)$  is at most  $1/6$ . But for  $\sigma < 1/6$ , we have
+      and by \Cref{Theorem: product of Dirichlet series converges} this series converges absolutely for  $\sigma > 1/6$ , so the
      abscissa of absolute convergence for  $\mathcal{L}_{\text{Qad } m}(s)$  is at most  $1/6$ . For the reverse inequality, choose any twist-minimal curve
      in the family and let  $n_0$  be its twist height. By \Cref{Theorem: relationship between twistL(s) and L(s)}(a), the
      coefficients of  $\mathcal{L}_{\text{Qad } m}(s)$  satisfy
+      \begin{equation}
+      \mathcal{L}_{\text{Qad } m}(n_0 c^6) \geq 2 \text{abs}(\mu(c))
+      \end{equation}
+      for every positive integer  $c$ . Hence the Dirichlet series for  $\mathcal{L}_{\text{Qad } m}(s)$  contains the divergent positive subseries

-       $\mathcal{L}_{\text{Qad } m}(\sigma) > \frac{2 \zeta(6s)}{\zeta(12s)}$ 
+       $2 n_0^{-\sigma} \sum_{c \geq 1} \frac{1}{c^{\sigma}} \text{abs}(\mu(c)) c^{6\sigma}$ 

-      by termwise comparison of coefficients, so the Dirichlet series for  $\mathcal{L}_{\text{Qad } m}(s)$  diverges when  $\sigma < 1/6$ , and (i) holds with  $\sigma_a = 1/6$ .
+      whenever  $\sigma \leq 1/6$ . Thus its abscissa of absolute convergence is at least  $1/6$ , and (i) holds with  $\sigma_a = 1/6$ .

-...  $\zeta(6s)/\zeta(12s)$  has a meromorphic contintuation to  $\sigma > 1/12$ , and so ...
+...  $\zeta(6s)/\zeta(12s)$  has a meromorphic continuation to  $\sigma > 1/12$ , and so ...

```

Submitted Thesis p. 191

... is absolutely convergent for $s > 1/12$, \Cref{Theorem: absolutely ...}
 +... is absolutely convergent for $\sigma > 1/12$, \Cref{Theorem: absolutely ...}

... $m(n)_{n \geq 1}$ is admissible with parameters $\left(1/6, 1/12, 13/42\right)$. We now apply \Cref{Theorem: Landau's Tauberian theorem} to ...
 +... $m(n)_{n \geq 1}$ is admissible with parameters $\left(1/6, 1/12, 13/84\right)$. We now apply \Cref{Theorem: Landau's Tauberian theorem} to ...

... most $O(X^{1/2} \deg B m \dots)$
 +... most $O(X^{1/(2 \deg B m)} \dots)$

Submitted Thesis p. 192

-The algorithm described in \cref{Section: Computations for $m = 7$ } can be adapted to ... $b_m) = m$ and $C(a_e, b_e) = e_0$, but also products of their conjugates.
 +The algorithm described in \cref{Section: Computations for $m = 7$ } can be adapted to enumerate elliptic curves admitting a cyclic m -isogeny for $m = 10, 25$. Doing so requires paying special attention to the primes 2 and 5 , rather than 3 and 7 , and requires writing $C_m(a, b) = e_0^2 n_0$, with n_0 squarefree. In step 3, if e_0^2 lies in the lookup range, we multiply representatives of norm e_0^2 by representatives of norm n_0 ; otherwise we square representatives of norm e_0 first. When $m = 10$, the lookup table in step 2 is restricted to pairs (a, b) with $a \geq b$; in step 3 we also use their conjugates. The bounded components at 2 and 5 are handled separately.

-For $m = 10$, running our algorithm out to $X = 10^{96}$ in Python took us approximately 17 CPU hours 61613.3902490139 seconds
 -on a single core, producing $106\,785\,277$ elliptic curves admitting a cyclic 10 -isogeny. To check the accuracy of our code, we confirmed that the j -invariants of these curves are distinct. For $X = 10^{96}$, we have
 +For $m = 10$, enumerating out to $X = 10^{96}$ produces $106\,785\,276$ elliptic curves admitting a cyclic 10 -isogeny. For $X = 10^{96}$, we have

-For $m = 25$, running our algorithm out to $X = 10^{138}$ took us approximately 10 CPU hours 36199.61623263359 seconds
 -on a single core, producing $34\,908\,299$ elliptic curves admitting a cyclic 25 -isogeny. To check the accuracy of our code, we confirmed that the j -invariants of these curves are distinct. For $X = 10^{138}$, we have
 +For $m = 25$, enumerating out to $X = 10^{138}$ produces $35\,007\,063$ elliptic curves admitting a cyclic 25 -isogeny. For $X = 10^{138}$, we have

Submitted Thesis p. 193

... $0.997\,115\dots$, \label{Equation: ...}
 +... $0.999\,937\dots$, \label{Equation: ...}

... compute $\text{twconst}\{10\}$ below. It ...
 +... compute $\text{twconst}\{25\}$ below. It ...

-We list the first few twist minimal elliptic curves admitting a cyclic 10 -isogeny ...
 +We list some twist minimal elliptic curves admitting a cyclic 10 -isogeny ...

-\{ E \in \text{twistE} with a cyclic 10 -isogeny ...
 +\{Some E \in \text{twistE} with a cyclic 10 -isogeny ...

Submitted Thesis p. 195

... 25 as a product over $p \leq Y$ gives us a lower bound
 +... 25 as a product over $p \leq Y$ gives the lower bound

Submitted Thesis p. 196

... appearing in \Cref{Theorem: asymptotic for $NQ(X)$ for $m = 10$...
 +... appearing in \Cref{Theorem: asymptotic for $twN(X)$ for $m = 10$...

... $\frac{13}{6} \cdot 63 \cdot 8 \cdot \prod_{\substack{p \dots \\ p \leq Y, \dots}}$
 +... $\prod_{\substack{p \dots \\ p \leq Y, \dots}}$

... $+\frac{2p}{p^6 \deg B m - 1} \cdot \frac{(p+1)}{(p^2 - p^{6 \deg B m})}$
 +... $+\frac{2p}{p^6 \deg B m - 1} \cdot \frac{(p+1)}{(p^2 - p^{6 \deg B m})}$

-for Q_m . The values $Q_m(2)$ and $Q_m(5)$ are recorded ... Recall \eqref{Equation: prime counting function}. Using Abel summation and Bennett--Martin--O'Bryant--Rechnitzer \cite[Theorem 1.4]{Bennett-Martin-OBryant-Rechnitzer}, we obtain
 +Put $\alpha_m = 6 \deg B m$ and $s_m = 2 - \alpha_m$. For $p > Y$, the nontrivial local increment is less than $2p^{-s_m} / (1 - Y^{-s_m})$. Thus, by Abel summation and Bennett--Martin--O'Bryant--Rechnitzer \cite[Theorem 1.4]{Bennett-Martin-OBryant-Rechnitzer}, for $Y \geq 8 \cdot 10^9$,

-\begin{aligned} \sum_{\substack{p > Y \\ p \equiv 1 \pmod{4}}} \frac{1}{p^2 + 1} &= -\frac{\pi(Y; 4, 1)}{Y^2 + \dots} + \log Y + \left(\frac{1}{Y} + \frac{5}{2 \log^2 Y} \right) \left(\frac{\pi}{Y} - \tan^{-1}(Y) \right). \\ Q_m &< Q_{\lfloor m \rfloor} \exp \left(\frac{Y^{1-s_m}}{(s_m-1) \log Y} \right) \end{aligned}

+\\parent{1+\\frac{5s_m}{2\\log Y}}.

Submitted Thesis p. 197

-In particular, letting $Y = 10^{11}$, we compute
+In particular, taking $Y = 10^{11}$ gives

-3.636\\,493\\,079\\,001\\,437\\,6 < \\Q {10} < 3.636\\,493\\,079\\,020\\,102, %52509.778564453125 seconds %15 CPU hours, $Y = 10^{11}$
+3.636\\,493\\,079\\,001\\,437\\,6 < \\Q {10} < 3.636\\,494\\,121\\,479\\,969

-4.244\\,853\\,881\\,138\\,272\\,6 < \\Q {25} < 4.244\\,853\\,881\\,160\\,06; %53808.07576751709 seconds %2 CPU hours, $Y = 10^{11}$
+4.244\\,853\\,881\\,138\\,272\\,6 < \\Q {25} < 4.244\\,853\\,894\\,726\\,220.

-these estimates require approximately \$15\$ CPU hours apiece.

... R_{10} , ...
+... $\text{calR}_{10}(1)$, ...

-1.141\\,059\\,04 \\cdot \\sqrt{\\frac{r_{10}(s_{10}) - r_{10}}{s_{10}^3}} < 1.6 \\cdot 10^{-8}.
+1.141\\,059\\,04 \\cdot \\sqrt{\\frac{r_{10}(s_{10}) - r_{10}}{s_{10}^3}}
+=1.516\\,12\\ldots \\cdot 10^{-6}.

... R_{25} , ...
+... $\text{calR}_{25}(1)$, ...

Submitted Thesis p. 198

-We therefore have
-\\begin{equation} \\begin{aligned} & \\twconst {10} \\approx 1.068\\,204 \\cdot \\text{with error bounded by} \\cdot 3.4 \\cdot 10^{-6}, \\ \\ & \\twconst {25} \\approx 0.754\\,252\\,0 \\cdot \\text{with error bounded by} \\cdot 9.6 \\cdot 10^{-7}. \\ \\ \\end{aligned} \\end{equation}

+Using the finite Euler products as point estimates, we therefore have
+\\begin{equation} \\begin{aligned} & \\twconst {10} \\approx 1.068\\,203\\,87 \\cdot \\text{with Monte Carlo standard error } 3.36 \\cdot 10^{-6}, \\ \\ & \\twconst {25} \\approx 0.754\\,251\\,990 \\cdot \\text{with Monte Carlo standard error } 9.58 \\cdot 10^{-7}. \\ \\ \\end{aligned} \\end{equation}

+The corresponding errors from truncating the Euler products are below $3.07 \\cdot 10^{-7}$ and $2.42 \\cdot 10^{-9}$, respectively.

-In this subsection, we estimate $\\constad {10} = 2 \\twistNad {10} (1/6) / \\zeta(2)$ and $\\constad {25} = 2 \\twistNad {10} (1/6) / \\zeta(2)$, the constants which appear in \\Cref{Theorem: asymptotic for NQ(X) for m = 10 and 25}, by computing the partial sums of $\\twistLad {10} (1/6)$ and $\\twistLad {25} (1/6)$:
+In this subsection, we estimate $\\constad {10} = 2 \\twistLad {10} (1/6) / \\zeta(2)$ and $\\constad {25} = 2 \\twistLad {25} (1/6) / \\zeta(2)$, the constants which appear in \\Cref{Theorem: asymptotic for NQ(X) for m = 10 and 25}, using the partial sums

... 0.869\\,838\\,621\\,652\\,207\\,3, ...
+... 0.411\\,595\\,409\\,503\\,702\\,919, ...

... 0.206\\,338\\,924\\,690\\,954\\,36.
+... 0.206\\,338\\,924\\,692\\,701\\,754.

Submitted Thesis p. 199

-If these bounds continue to hold for larger Y , then \\begin{equation} \\begin{aligned} \\sum_{n > ...} \\constad {25} & \\approx 0.25087816 \\cdot \\text{with error bounded by} \\cdot 1.2 \\cdot 10^{-7}. \\ \\ \\end{aligned} \\end{equation}

+If these bounds continue to hold for larger Y , partial summation gives
+\\begin{equation} \\begin{aligned} & \\sum_{n > Y} \\frac{\\twisthad m(n)}{n^{1/6}} \\ \\ & + -Y^{-1/6} \\twistNad m(Y) + \\frac{1}{6} \\int_Y^{\\infty} \\twistNad m(x) x^{-7/6} dx. \\ \\ \\end{aligned} \\end{equation}

+Using $\\twistNad {10} (10^{96}) = 106\\,785\\,276$ and $\\twistNad {25} (10^{138}) = 35\\,007\\,063$, the two tails are respectively less than $1.12266724 \\cdot 10^{-8}$ and $2.83347 \\cdot 10^{-16}$. Hence
+\\begin{equation} \\begin{aligned} & \\twistLad {10} (1/6) \\approx 0.411595415117039 \\cdot \\text{with conditional error } < 5.62 \\cdot 10^{-9}, \\ \\ & \\twistLad {25} (1/6) \\approx 0.20633892469270190 \\cdot \\text{with conditional error } < 1.42 \\cdot 10^{-16}, \\ \\ \\end{aligned} \\end{equation}

+and therefore
+\\begin{equation} \\begin{aligned} & \\constad {10} \\approx 0.500440015697013 \\cdot \\text{with conditional error } < 6.83 \\cdot 10^{-9}, \\ \\ & \\constad {25} \\approx 0.25087804897622103 \\cdot \\text{with conditional error } < 1.73 \\cdot 10^{-16}. \\ \\ \\end{aligned} \\end{equation}

Submitted Thesis p. 200

... 13\$, with an improved error term. ...
+... 13\$.

-... theoretical estimate
+... theoretical estimate.

Submitted Thesis p. 202

- $\mathbb{Z} = 2(a^6 + 512a^5b - 13073a^4b^2 + 34860a^3b^3 - 157099a^2b^4 + 211330ab^5 - 655108b^6)$.
+ $\mathbb{Z} = 2(a^6 + 512a^5b - 13073a^4b^2 + 34860a^3b^3 - 157099a^2b^4 + 211330ab^5 - 655108b^6)$.

-... b) $\{C_{13}\}$...
+... b) $\{C_{II}^{13}\}$...

-... $\mathbb{Z}$ -groomed, let ℓ be prime, and let $v \in \mathbb{Z}_{\geq 0}$. Then the following statements hold.
+... $\mathbb{Z}$ -groomed, let ℓ be prime, and let $v \in \mathbb{Z}_{>0}$. Then the following statements hold.

Submitted Thesis p. 203

-... and 25} however.
+... and 25}, however.

-... the minimality defect $\text{\eqref{Equation: ...}}$
+... the twist minimality defect $\text{\eqref{Equation: twist ...}}$

Submitted Thesis p. 204

-... $\{13\}(t,1) = \text{\Res}(tA \dots)$
+... $\{13\}(t,1) = \text{\Res}(t\mathbf{f} \dots)$

-This proves (a)
+This proves (a).

-... $\{13\}(b,0) \dots$
+... $\{13\}(a,0) \dots$

Submitted Thesis p. 205

-... 3 \mid \mathfrak{h}_{III}^{13}(a, \dots)
+... 3 \mid \mathfrak{C}_{III}^{13}(a, \dots)

-... 13 lesser} is ...
+... 13} are ...

Submitted Thesis p. 206

-... $e_1 = 3^{v_1} 13^{w_1} \dots$
+... $e_1 = 2^{u_1} 3^{v_1} 13^{w_1} \dots$

-... $13^{w_2} e_2 \text{\prm}$, where $\gcd(e_1 \text{\prm}, \dots)$
+... $3^{v_2} 13^{w_2} e_2 \text{\prm}$, where $\gcd(e_1 \text{\prm}, 2 \cdot \dots)$

-... $\gcd(e_2 \text{\prm}, 2 \cdot 13) \dots$
+... $\gcd(e_2 \text{\prm}, 2 \cdot 3 \cdot 13) \dots$

-... $(a, b) \text{\psmod} \{(e_2 \text{\prm})^2\}$,
+... $(a, b) \text{\equiv} 0 \text{\psmod} \{(e_2 \text{\prm})^2\}$,

-... $w_2 > 0$ then $A_{13}(a, b) \text{\equiv} 0 \text{\psmod} \{13^{2w_1}\}$ and $B_{13}(a, b) \text{\equiv} 0 \text{\psmod} \{13^{3w_1}\}$, ...
+... $w_2 > 0$ then $A_{13}(a, b) \text{\equiv} 0 \text{\psmod} \{13^{2w_2}\}$ and $B_{13}(a, b) \text{\equiv} 0 \text{\psmod} \{13^{3w_2}\}$, ...

-... $\text{\psmod} \{13^{3w_1}\}$, but $C_{II} \dots$
+... $\text{\psmod} \{13^{3w_2}\}$, but $C_{II} \dots$

-... $\not\equiv 0 \text{\psmod} \{13\}$.
+... $\equiv 0 \text{\psmod} \{13^2\}$.

Submitted Thesis p. 207

-\item For all prime $\ell \neq 2$, ...
+\item For every prime $\ell \neq 2$, ...

Submitted Thesis p. 208

```

-... = 0(2^{\omega(e_1} ...
+... = 0(2^{\omega(e_1} ...

-      For parts (a) and (b), by the CRT (Sun Zi theorem), it suffices to consider  $e = \ell^v$  a power of a prime. For  $\ell \neq 2, 3, 13$ , both claims follow from \Cref{Lemma: 2 and 3 and 13 are the ungroomed primes for  $m = 13$ }(a)–(b). But a finite computation verifies our claim in these cases as well (see the proof of (e) below).
+      For parts (a) and (b), by the CRT (Sun Zi theorem), it suffices to consider  $e = \ell^v$  a power of a prime. For  $\ell \neq 2, 3, 13$ , both claims follow from \Cref{Lemma: 2 and 3 and 13 are the ungroomed primes for  $m = 13$ }(a)–(b). For the exceptional primes, the stated conditions follow directly from \Cref{Definition T(e) for  $m = 13$ } and \eqref{Equation: twist defect powers}, apart from the two assertions that no pairs are permitted when  $2 \mid e_1$  or  $3 \mid e_2$ . The first follows by checking the defining conditions modulo  $8$ , and the second by checking them modulo  $27$ . Part (b) then follows from the divisibility conditions in part (a).

-We observe  $A(1, 0) = -3$  and  $B(1, 0) = 2$  are ...
+We observe  $A_{13}(1, 0) = -3$  and  $B_{13}(1, 0) = 2$  are ...

-... as  $A(1, 0) = -3$  and  $B(1, \dots$ 
+... as  $A_{13}(1, 0) = -3$  and  $B_{13}(1, \dots$ 

```

Submitted Thesis pp. 208–209 (deliberate completion)

```

+      For  $v \geq 1$ , the  $13$ -adic values are
+      \begin{equation}\label{Equation: cT13 at 13}
+      \begin{cases}
+      1+12 \cdot 13^{3v-2}, & 1 \leq v \leq 3, \\
+      1+13^8, & v \geq 4,
+      \end{cases}
+      \quad
+      \begin{cases}
+      cT_{13}(1, 13^v) =
+      \begin{cases}
+      1+13^{2v-2}, & 1 \leq v \leq 3, \\
+      1+13^5, & v \geq 4.
+      \end{cases}
+      \end{cases}
+      \end{equation}
+
+It remains to treat  $\ell=13$ . Modulo  $13$  we have
+\begin{equation}
+\begin{aligned}
+&f_{13}(t) \equiv -3(t-2)(t-3)(t+3)^6, \\
+&g_{13}(t) \equiv 2(t-2)(t-3)^2(t+3)^9.
+ \end{aligned}
+\end{equation}
+Thus only the residue classes  $t \equiv 2, 3, 10 \pmod{13}$  can contribute. At  $t \equiv 2$ , the factor  $h_{II}$  has a simple root while  $h_{III}$ ,  $t$ , and  $g$  are units, so Hensel's lemma gives exactly one class in  $\mathcal{T}_{13}(13^v, 1)$  for every  $v$ . Likewise, the simple root  $t \equiv 3$  of  $h_{III}$  gives exactly one class in  $\mathcal{T}_{13}(1, 13^v)$  for every  $v$ .
+
+For the common residue class, write  $t=10+13u$ . Then
+\begin{equation}
+\begin{aligned}
+&h_{II}(t) \equiv 13H_{II}(u), \\
+&H_{II}(u) \equiv 13u^2+21u+9, \\
+&h_{III}(t) \equiv 13H_{III}(u), \\
+&H_{III}(u) \equiv 13u^2+20u+8, \\
+&t \equiv 13^4F(u), \\
+&F(u) \equiv -3(u^4-15u^3-31u^2-20u-4), \\
+&tg(u) \equiv 13^6G(u), \\
+&G(u) \equiv 2(u^6+44u^5+83u^4+20u^3-59u^2-50u-12).
+ \end{aligned}
+\end{equation}
+Modulo  $13$  these satisfy
+\begin{equation}
+H_{II}(u) \equiv 8(u-7), \quad \text{quad} \\
+H_{III}(u) \equiv 7(u-10), \quad \text{quad} \\
+F(u) \equiv -3(u-7)^4, \quad \text{quad} \\
+G(u) \equiv 2(u-10)^6.
+\end{equation}
+Consequently, for  $\mathcal{T}_{13}(13^v, 1)$  the defining convention forces  $H_{III}(u)$  to be a unit. If  $v \leq 3$ , the displayed powers  $13^6$  and  $13^9$  already supply the required divisibility, giving  $(13-1)13^{3v-2}$  classes on this branch. If  $v \geq 4$ , then  $G(u)$  is a unit and the condition on  $g$  is exactly  $13^{3v-9} \mid H_{II}(u)$ . The latter polynomial has a simple root modulo  $13$ , so there are exactly  $13^8$  such classes modulo  $13^{3v}$ . Including the branch through  $t \equiv 2$  gives the first formula of \eqref{Equation: cT13 at 13}.
+
+For  $\mathcal{T}_{13}(1, 13^v)$ , the common branch must instead satisfy  $13 \mid H_{III}(u)$ , while  $H_{II}(u)$  and  $F(u)$  are units. A further substitution  $u=10+13w$  gives
+\begin{equation}
+13^{-1}G(10+13w) \equiv -5 \pmod{13},
+\end{equation}
+so  $v_{13}(G(u))=1$  throughout this branch. If  $r=v_{13}(H_{III}(u)) \geq 1$ , the two divisibility conditions are therefore

```

```

+\begin{equation}
+6+r\geq2v,\qquad 10+2r\geq3v.
+\end{equation}
+For  $v\leq3$  these impose only  $r\geq1$ , giving  $13^{2v-2}$  classes modulo  $13^{2v}$ ; for  $v\geq4$  they reduce to  $r\geq2v-6$ ,
giving  $13^5$  classes. Including the branch through  $t\equiv3$  gives the second formula of \eqref{Equation: cT13 at 13}.
-
-Notably, \Cref{Lemma: bound on T(e)} for  $m = 13$  does not furnish the values of  $cT_{13}(13^v, 1)$  and  $cT_{13}(1, 13^v)$  for  $v \geq 1$ . By Hensel's Lemma, these functions are constant for sufficiently large  $v$ , but our somewhat naive code runs into
memory issues before verifying these plateaued values. Unfortunately, this obstructs the computation of  $Q_{13}$  below;
however, it poses no issue for our theoretical results.

```

Submitted Thesis p. 209

```

-...  $h_{13}(t)$  ...
+... the corresponding polynomial ...

-...  $(2^{\ell_1}, 1) = 0$  for  $\ell_1$  ...
+...  $(2^v, 1) = 0$  for  $v$  ...

```

Submitted Thesis p. 210

```

-}{All nonzero  $cT_{13}(3^v, 1)$ }
+}{All nonzero  $cT_{13}(1, 2^v)$ }

-... 7} and \Cref{Theorem: asymptotic for NQ(X)} for  $m = 10$  and ...
+... 7} and \Cref{Theorem: Controlling size of twist minimality defect} for  $m = 5$  and  $10$  and ...

-... {II, 13}  $C_m(a, \dots)$ 
+... {II, 13}  $C_{II_{13}}(a, \dots)$ 

-... {III, 13}  $C_m(a, \dots)$ 
+... {III, 13}  $C_{III_{13}}(a, \dots)$ 

```

Submitted Thesis p. 211

```

-... If  $C_{II_{13}}(a, b) = e_{\textup{II}}^3 n_{\textup{II}}$ , with  $n_0$  cube-free, and  $C_{III_{13}}(a, b) = e_{\textup{III}}^2 \dots$ 
+... If  $C_{II_{13}}(a, b) = e_{\textup{II}}^3 n_{\textup{II}}$ , with  $n_{\textup{II}}$  cube-free, and  $C_{III_{13}}(a, b) = e_{\textup{III}}^2 \dots$ 

-... evaluating  $C_{III}$  ...
+... evaluating  $C_{II}$  ...

```

Submitted Thesis p. 213

```

-... cases,  $27 |\mathbf{B}_m(a, b)|^2 > 4 |\mathbf{A}_m(a, b)|^3$ . For ...
+... cases,  $27 |\mathbf{B}_{13}(a, b)|^2 > 4 |\mathbf{A}_{13}(a, b)|^3$ . For ...

-... Write  $C_{II_{13}}(a, b) = e_{\textup{II}}^3 n_{\textup{II}}$ , with  $n_0$  cube-free, and  $C_{III_{13}}(a, b) = e_{\textup{III}}^2 \dots$ 
+... Write  $C_{II_{13}}(a, b) = e_{\textup{II}}^3 n_{\textup{II}}$ , with  $n_{\textup{II}}$  cube-free, and  $C_{III_{13}}(a, b) = e_{\textup{III}}^2 \dots$ 

-...  $13(a)$  and , ...
+...  $13(a)$ , ...

-...  $u = v = w = \dots$ 
+...  $u, v, w \leq \dots$ 

```

Submitted Thesis p. 214

```

-...  $b + 1$ :
+...  $b^2 + 1$ .

-we maximize ...
+We maximize ...

-... minimize ...
+... minimize ...

-... analogues to \Cref{Theorem: Controlling size of twist minimality defect} for  $m = 13(b)$ .
+... analogous bounds.

-... if  $C_{II_{13}}(a, b) = e_{\textup{II}}^3 n_{\textup{II}}$ , with  $n_0$  cube-free, and  $C_{III_{13}}(a, b) = e_{\textup{III}}^2 \dots$ 

```

```
+... if $\CII {13} (a, b) = e_{\textup{II}}^3 n_{\textup{II}}$, with $n_{\textup{II}}$ cube-free, and $\CIII {13} (a, b) = e_{\textup{III}}^2 \dots$

-... 7^6} ...
+... 13^6} ...

-... \upperratio {13} ...
+... \upperratio {11, 13} ...
```

Submitted Thesis p. 215

```
-... X^{\frac{1}{18}}\{d^2\} ...
+... X^{\frac{1}{12}}\{d^2\} ...

-... 0\parent{\frac{2^{\omega(e)}}{X^{\frac{1}{24}}}\{d e^{\frac{3}{2}}\}}
+... 0\parent{\frac{2^{2 \omega(e)}}{X^{\frac{1}{24}}}\{d\}}
```

Submitted Thesis p. 216

```
-... $X \geq 2$ and $d, e \geq 1$.
+... $X \geq 2$ and $e \geq 1$.

-... \ $a/b$ ...
+... \ $(a:b)$ ...
```

Submitted Thesis p. 217

```
-      = & \R 7 \varphi(e_1^3 e_2^2) X^{\frac{1}{12}} \sum_{\substack{e_1 e_2 = e \\ \gcd(e_1, e_2) = 1}} \frac{\cT {13}(e_1, e_2)}{e_1^6 e_2^4} \backslash \backslash
- &+ 0\parent{\varphi(e_1^3 e_2^2) X^{\frac{1}{24}} \sum_{\substack{e_1 e_2 = e \\ \gcd(e_1, e_2) = 1}} \frac{\cT {13}(e_1, e_2)}{e_1^3 e_2^2} } \backslash \backslash
+      = & \R {13} X^{\frac{1}{12}} \sum_{\substack{e_1 e_2 = e \\ \gcd(e_1, e_2) = 1}} \frac{\varphi(e_1^3 e_2^2) \cT {13}(e_1, e_2)}{e_1^6 e_2^4} \backslash \backslash
+ &+ 0\parent{X^{\frac{1}{24}} \sum_{\substack{e_1 e_2 = e \\ \gcd(e_1, e_2) = 1}} \frac{\varphi(e_1^3 e_2^2) \cT {13}(e_1, e_2)}{e_1^3 e_2^2} } \backslash \backslash
```

Submitted Thesis p. 218

```
-... {13}(x; ...
+... {13}(X; ...

-... \frac{\R {13} X^{\frac{1}{12}}}{\prod_{\ell \mid e} \parent{1 - \frac{1}{\ell}}}} ...
+... \R {13} X^{\frac{1}{12}} \prod_{\ell \mid e} \parent{1 - \frac{1}{\ell}} ...

-\cM {13}(x;e)
+\cM {13}(X;e)
```

Submitted Thesis p. 219

```
-      \calR {13}(1) = \set{(a, b) \in \bbR^2 : \rawheight(\A m (a, b), \B m (a, b)) \leq 1, b \geq 0}.
+      \calR {13}(1) = \set{(a, b) \in \bbR^2 : \rawheight(\A {13} (a, b), \B {13} (a, b)) \leq 1, b \geq 0}.
```

Submitted Thesis p. 220

```
-... = \twconst {13} X^{\frac{1}{12}} + 0\parent{\max\parent{\frac{X^{\frac{1}{12}} \log^3 y}{y}, X^{\frac{1}{24}} y^{\frac{5}{4}} \log X \log^7 y}}
+... = \twconst {13} X^{\frac{1}{12}} + 0\parent{\max\parent{\frac{X^{\frac{1}{12}} \log^3 y}{y^{\frac{1}{2}}}, X^{\frac{1}{24}} y^{\frac{5}{4}} \log X \log^7 y}}

-... e^{\frac{1}{6}} ...
+... e^{\frac{1}{4}} ...
```

Submitted Thesis p. 221

```
-... 0\parent{\frac{X^{\frac{1}{24}}}{...}
+... 0\parent{\frac{X^{\frac{1}{12}}}{...}}
```

Submitted Thesis p. 222

```

... &\colonequals \sum_{a \geq 0} \frac{\varphi_{1/2}(p^a)}{1} ...
... &\colonequals 1 + \sum_{a \geq 1} \frac{\varphi_{1/2}(p^a)}{1} ...

... {13}.
... {12}.

-      We have been unable to compute  $Q_{13}(13)$  because we do not know  $cT_{13}(13^v, 1)$  and  $cT_{13}(1, 13^v)$  for all  $v \geq 0$ . The square roots appear in \eqref{Equation: Euler factors for m = 13} because of the generalized Jordan totient function  $\varphi_{1/2}$ . For instance, for  $p = 3$  we have
+      \frac{79 \cdot 239 \cdot 719 + 73 \cdot 639 \cdot 763 \sqrt{13}}{79 \cdot 270 \cdot 842}, & \text{if } p=13; \setminus
+      For  $p=13$ , substituting \eqref{Equation: cT13 at 13} into the defining series and summing the geometric tails gives
+      \begin{equation}
+      Q_{13}(13) = \frac{79 \cdot 239 \cdot 719 + 73 \cdot 639 \cdot 763 \sqrt{13}}{79 \cdot 270 \cdot 842} = 4.349 \cdot 034 \cdot 925 \cdot 218 \cdot 983 \cdot 880 \ldots
+      \end{equation}
+      The square roots in \eqref{Equation: Euler factors for m = 13} appear because of the generalized Jordan totient function  $\varphi_{1/2}$ . For instance, for  $p = 3$  we have

... + 7 \sqrt{3}^{36}.
... + 17 \sqrt{3}^{36}.

```

Submitted Thesis p. 223

```

-0\parent{X^{-1/12} \log X \sum_{n \leq y} 4^{-\omega(n)} ...
+0\parent{X^{-1/24} \log X \sum_{n \leq y} 4^{-\omega(n)} ...

-      We have
-      \begin{equation}
-      \text{twistNadgy}_{13}(X) = 0\parent{\frac{X^{-1/12} \log^3 y}{y^{1/2}}}
-      \end{equation}
-      for  $X, y \geq 2$ .
+For every  $\epsilon > 0$ , we have
+\begin{equation}
+\text{twistNadgy}_{13}(X) = O_{\epsilon} \parent{\frac{X^{-1/12+\epsilon}}{y^{1/6}}} + X^{-1/14+\epsilon}
+\end{equation}
+for  $X, y \geq 2$ .

```

Submitted Thesis p. 224

```

-By \Cref{Lemma: bound on T(e) for m = 13},  $cT_{13}(e_1, e_2) = \dots$  of  $f/n^{\sigma}$ , we conclude \begin{equation} \text{twistNadgy}_m(X) =
0\parent{\frac{X^{-1/12} \log^3 y}{y^{1/2}}} \end{equation} as desired.
+We have  $\text{CII}_{13} = \mathcal{C} 7$  and  $\text{CIII}_{13} = \mathcal{C} 25$ . The norm-form lattice arguments of \Cref{Lemma: asymptotic for M(X; e) for m = 7,
Lemma: asymptotic for M(X; e) for m = 5 and 10 and 25}, together with the Principle of Lipschitz, give
+\begin{equation}
+\begin{aligned}
+ \# \{(a, b): e^3 \mid \text{CII}_{13}(a, b) \} & \text{ for some } e > E \\
+ \# \{(a, b): e^2 \mid \text{CIII}_{13}(a, b) \} & \text{ for some } e > E \\
+ \# \{(a, b): e^2 \mid \text{CIII}_{13}(a, b) \} & \text{ for some } e > E \\
+ \# \{(a, b): e^2 \mid \text{CIII}_{13}(a, b) \} & \text{ for some } e > E \\
+ \end{aligned}
+\end{equation}
+for primitive pairs with  $|a|, |b| \leq B$ ; here the exceptional  $3$ - and  $2$ -parts are bounded, since  $v_3(\text{CII}_{13}) \leq 3$  and  $v_2(\text{CIII}_{13}) \leq 3$  for primitive pairs.

+Let  $D = e_{\text{II}} e_{\text{III}} e'$  be the actual twist minimality defect, with  $e' \mid 78$ , as in \Cref{Theorem: Controlling size of twist minimality defect for m = 13}(b). By \Cref{Proposition: bounding the summands of twN calE X}, the large-defect sum is bounded, up to  $\tau(D)^2$ , by pairs of twist height at most  $X$  with  $D > y$ . The same theorem gives  $|a|, |b| \ll X^{1/14}$  and  $D \ll X^{5/42}$ , so  $\tau(D)^2 \ll \epsilon X^{1/42}$ , so  $\tau(D)^2 \ll \epsilon X^{1/42}$  proves the lemma.

+On a dyadic interval  $D_0 \leq D < 2D_0$ , we have  $|a|, |b| \ll X^{1/24} D_0^{-1/4}$  and either  $e_{\text{II}} \mid D_0^{1/3}$  or  $e_{\text{III}} \mid D_0^{2/3}$ . The preceding estimates therefore give
+\begin{equation}
+O_{\epsilon} \parent{X^{-1/12+\epsilon} D_0^{-1/6} + X^{1/24+\epsilon} D_0^{-1/4}}.
+\end{equation}
+Summing dyadically for  $D_0 > y$  and using  $D_0 \ll X^{5/42}$  proves the lemma.

... 13, which we restate here with an improved error term in the notations we ...
... 13, which we restate here in the notations we ...

-      We have
+      For every  $\epsilon > 0$ , we have

-      \text{twistNad}_{13}(X) = \text{twistNad}_{13}(X) = \text{twconst}_{13} X^{-1/12} + O\parent{X^{7/108} \log^{43/9} X}
+      \text{twistNeq}_{13}(X) = \text{twistNad}_{13}(X) = \text{twconst}_{13} X^{-1/12} + O_{\epsilon} \parent{X^{4/51+\epsilon}}

-Let  $\delta$  be a positive quantity with  $\delta \ll X^{-1/24}$ . \textit{A fortiori}, we have ... = \text{twconst}_m X^{-1/\deg B} + O\parent{X^{7/108} \log^{43/9} X} \end{equation} as desired.
+Combining \Cref{Lemma: asymptotic for twN <= y(X) for m = 13, Lemma: bound on twN > y(X) for m = 13} and taking  $\delta = X^{-1/34}$ , we have
+\begin{equation}

```

+ $X^{\frac{1}{24}}y^{\frac{5}{4}}=X^{\frac{1}{12}}y^{-\frac{1}{6}}=X^{\frac{4}{51}}$.
+ $\end{equation}$
+The remaining error terms are smaller, and the logarithmic factors are absorbed into X^{ϵ} .

Submitted Thesis p. 225

-... $7/108$
+... $4/51$
- $\mu_{\text{twistLad } \{13\}}(\sigma) < 13/84$
+ $\mu_{\text{twistLad } \{13\}}(\sigma) \leq \max\{0, 17-204\sigma\}$

Submitted Thesis p. 226

-... $7/108$.
+... $4/51$.
- The proof is structurally identical to the one given for $\text{Cref}\{\text{Corollary: twL}(s) \text{ has a meromorphic continuation for } m = 7\}$.
+Abel summation applied to $\text{Cref}\{\text{Theorem: asymptotic for twN}(X) \text{ for } m = 13\}$ gives (a)–(b). On fixed lines $4/51 < \sigma_0 < 1/12$, the remainder integral is $O(1+|t|)$, while absolute convergence gives vertical-growth exponent O for $\sigma > 1/12$; subconvexity of the μ -function and $\text{Cref}\{\text{Theorem: muzeta}(\sigma)\}$ then give (c).
-... analytic behavior of $L_{\text{Qad } m}(s)$ for $m = 10, 25$.
+... analytic behavior of $L_{\text{Qad } \{13\}}(s)$.

Submitted Thesis p. 227

-The sequence $\text{parent}\{\text{hQad } m(n)\}_{n \geq 1}$ is admissible ...
+The sequence $\text{parent}\{\text{hQad } \{13\}(n)\}_{n \geq 1}$ is admissible ...
-... admissible for $m = 10$ and 25 , the sequence $\text{parent}\{\text{hQad } \{13\}(n)\}_{n \geq 1}$...
+... admissible for $m = 13$, the sequence $\text{parent}\{\text{hQad } \{13\}(n)\}_{n \geq 1}$...
-... $\{13\}(n)\}_{n \geq 1}$ is admissible with parameters $\text{parent}\{1/6, 1/12, 13/42\}$. We now apply $\text{Cref}\{\text{Theorem: Landau's Tauberian theorem}\}$ to ...
+... $\{13\}(n)\}_{n \geq 1}$ is admissible with parameters $\text{parent}\{1/6, 1/12, 13/84\}$. We now apply $\text{Cref}\{\text{Theorem: Landau's Tauberian theorem}\}$ to ...
-We suspect that the true error on $N_{\text{Qad } m}(X)$ is at most $O(X^{\frac{1}{12} + \epsilon})$, and the true error on $\text{twistNad } \{13\}$...
+We suspect that the true error on $N_{\text{Qad } \{13\}}(X)$ is at most $O(X^{\frac{1}{12} + \epsilon})$, and the true error on $\text{twistNad } \{13\}$...

Submitted Thesis p. 228

-... abiding by its notations, ...
+... using its notation, ...

Submitted Thesis pp. 228–231 (deliberate completion)

-In this section, we describe an algorithm to enumerate the elliptic curves with a cyclic 13 -isogeny and twist height at most X .
We then use the list of elliptic curves admitting a cyclic 13 -isogeny to estimate $\text{constad } 13$. However, our ignorance about $\text{cT } \{13\}(13^v, 1)$ and $\text{cT } \{13\}(1, 13^v)$ prevents us from computing $\text{Q } \{13\}(13)$, and thus from computing $\text{twconst } \{13\}$.
+In this section, we describe an algorithm to enumerate the elliptic curves with a cyclic 13 -isogeny and twist height at most X .
We then numerically evaluate the Euler product $\text{Q } \{13\}$ and the twist-counting constant $\text{twconst } \{13\}$, and use the list of elliptic curves admitting a cyclic 13 -isogeny to estimate $\text{constad } 13$.
+
+subsection*{Computing $\text{twconst } \{13\}$ }
+
+We first estimate the Euler product $\text{Q } \{13\}$. For $p \notin \{2, 3, 13\}$, set
+
$$\text{A}_p \colonequals \frac{2p^{\frac{1}{2}} + (p+1)(p^{\frac{1}{2}}+1)}{(p+1)(p^2+p^{\frac{3}{2}}+p+p^{\frac{1}{2}}+1)},$$

+
$$\text{B}_p \colonequals \frac{2p^{\frac{1}{2}} + (p+1)(p^2+p^{\frac{3}{2}}+p+p^{\frac{1}{2}}+1)}{(p+1)(p^2+p^{\frac{3}{2}}+2p^2+p^{\frac{1}{2}}+2)}.$$

+The first case of $\text{Equation: Euler factors for } m = 13$ satisfies
+
$$\text{A}_p^2 \left(p^2 + p^{\frac{3}{2}} + 2p^2 + p^{\frac{1}{2}} + 2 \right) = (p+1) \left(p + p^{\frac{1}{2}} + 1 \right) \left(p^2 + p^{\frac{3}{2}} + p + p^{\frac{1}{2}} + 1 \right).$$

+Consequently, away from the three exceptional primes, the nontrivial local increment is A_p for $p \equiv 1 \pmod{4}$, together with B_p for $p \equiv 3 \pmod{4}$. In particular,
+
$$0 < \text{A}_p < 2p^{-\frac{3}{2}}, \quad \text{A}_p \ll 2p^{-\frac{5}{2}}.$$

+Let $Q_{\{13,\leq Y\}}$ denote the Euler product truncated at $p\leq Y$. Since $\log(1+x)<x$, for $Y\geq 8\cdot 10^9$ the same Abel-summation argument using the explicit prime-counting estimates of Bennett--Martin--O'Bryant--Rechnitzer \cite[Theorem 1.4]{Bennett-Martin-OBryant-Rechnitzer} gives

$$\log\frac{Q_{\{13,\leq Y\}}}{Y^{-1/2}\log Y}\leq \frac{1}{4}\log Y + \frac{1}{4}\log Y + \frac{1}{4}\log Y + \frac{1}{4}\log Y.$$

+At $Y=8\cdot 10^9$, direct computation gives

$$Q_{\{13,\leq Y\}}=9.775\,286\,929\,620\,052\,484\ldots$$

+while the exponent in the preceding upper bound is less than $1.141\,882\cdot 10^{-6}$. Thus

$$9.775\,286\,929\,620\,052 < Q_{\{13\}} < 9.775\,298\,091\,844\,354.$$

+Using the rejection-sampling estimate for $R_{\{13\}}$ below and the finite Euler product as a point estimate, we obtain

$$\frac{Q_{\{13\}}}{R_{\{13\}}^{\zeta(2)}}\approx 0.968\,679\,323.$$

+The Monte Carlo standard error propagated from $R_{\{13\}}$ is approximately $3.58\cdot 10^{-6}$, while the deterministic uncertainty from truncating the Euler product is less than $1.11\cdot 10^{-6}$. We report these separately, since the former is a statistical standard error rather than an absolute error bound.

Submitted Thesis p. 229

... $b > 0$, we ...
+... $b > 0$ and $A_{\{13\}}(a,b)^{3+27}B_{\{13\}}(a,b)^2 \neq 0$, we ...

Submitted Thesis p. 231

... $2\,\text{twistNad}_{\{13\}}(1/6)/\zeta(2)$...
+... $2\,\text{twistLad}_{\{13\}}(1/6)/\zeta(2)$...

... $0.680\,532\,123\,1018\,161$.
+... $0.680\,532\,123\,101\,821\,172$.

-Ironically, although we have an estimate for $\text{constad}_{\{13\}}$, we are unable to estimate ... $\text{colonequals } 33\,570\,382\,383$ of our first $s_{\{13\}} \text{colonequals } 61\,749\,000\,000$ samples lie in $R_{\{13\}}$, so

+pagebreak

+We estimate $R_{\{13\}}$ by rejection sampling on $[-0.8228, 0.8228] \times [0, 0.1822]$. Of $s_{\{13\}} \text{colonequals } 61\,749\,000\,000$ samples, $r_{\{13\}} \text{colonequals } 33\,570\,382\,383$ lie in $R_{\{13\}}$, giving

Submitted Thesis p. 232

... $8, 9, 12, 16, 18\}$ in $4, 6, 8, 9, 12, 16, 18\}$ \label{Chapter: ...
+... $8, 9, 12, 16, 18\}$ in $6, 8, 9, 12, 16, 18\}$ \label{Chapter: ...

Submitted Thesis p. 234

... $12, 15, \dots$
+... $12, 16, \dots$

Submitted Thesis p. 235

... $\mid g_m(t)$...
+... $\mid f_m(t)$...

Submitted Thesis p. 236

-\$16\$ & 3^2 & 3^5 & 3^8 & 3^{11} & --- \\
+\$16\$ & --- & --- & --- & --- & --- \\

Submitted Thesis p. 237

```
-... b)$ may ...
+... b), \B m (a, b))$ may be ...

-... = $ ...
+... = 4$ ...
```

Submitted Thesis p. 238

```
-\begin{lemma}\label{Lemma: fundamental sieve for for m ...
+\begin{lemma}\label{Lemma: fundamental sieve for m ...

-\twNad 4 (X) = ...
+\twNeq 4 (X) = ...

-... 3} \sum_{e \mid n} \mu(n/e) \cM m (e^6 X; n).
+... 3} \sum_{e \mid n} \mu(n/e) \cM 4 (e^6 X; n).
```

Submitted Thesis p. 239

```
-... m(X; e) = \sum_{\substack{d \mid 11 X^{1/2} \deg B m} \setminus \gcd(d, e) = ...
+... m(X; e) = \sum_{\substack{d \mid 11 X^{1/(2 \deg B m)} \setminus \gcd(d, e) = ...

-... X^{1/2 \deg B m}\{d\}
+... X^{1/(2 \deg B m)}\{d\}

-... 0\parent{2^{\omega(e)} X^{1/2 \deg B m} ...
+... 0\parent{2^{\omega(e)} X^{1/(2 \deg B m)} ...

- for $X \geq 2$ and $d, e \geq 1$.
+ for $X \geq 2$ and $e \geq 1$.

-... however, because \Cref{Proposition: bounding ...
+... however, because by \Cref{Proposition: bounding ...
```

Submitted Thesis p. 240

```
-... 3^4} ...
+... 3^5} ...
```

Submitted Thesis p. 241

```
-... \Cref{Lemma: fundamental sieve for for m of genus $0$} ...
+... \Cref{Lemma: fundamental sieve for m of genus $0$} ...

-... \twconsteq m X^{1/\deg B m} + 0\parent{X^{1/2 \deg B m} \log X},
+... \twconsteq m X^{1/\deg B m} + 0\parent{X^{1/(2 \deg B m)} \log X},

-... \Cref{Lemma: fundamental sieve for for m of genus $0$}, ...
+... \Cref{Lemma: fundamental sieve for m of genus $0$}, ...
```

Submitted Thesis p. 242

```
-... 0\parent{2^{\omega(n)} e^{3/\deg B m} X^{1/2 \deg B m} \log e^6 X}.
+... 0\parent{2^{\omega(n)} e^{3/\deg B m} X^{1/(2 \deg B m)} \log e^6 X}.

-... and \eqref{Equation: twconst for ...
+... and \eqref{Equation: twconsteq for ...
```

Submitted Thesis p. 243

```
- \sum_{n \mid 2^4 \cdot 3^5} \sum_{e \mid n} \mu\parent{n/e} 0\parent{2^{\omega(n)} e^{3/\deg B m} X^{1/2 \deg B m} \log e^6 X} \setminus
- =& 0\parent{X^{1/2 \deg B m} \log X \sum_{n \mid 2^4 \cdot 3^5} 2^{\omega(n)} \sum_{e \mid n} \abs{\mu\parent{n/e}} e^{3/\deg B m}}
- =& 0\parent{X^{1/2 \deg B m} \log X},
+ \sum_{n \mid 2^4 \cdot 3^5} \sum_{e \mid n} \mu\parent{n/e} 0\parent{2^{\omega(n)} e^{3/\deg B m} X^{1/(2 \deg B m)} \log e^6 X} \setminus
+ =& 0\parent{X^{1/(2 \deg B m)} \log X \sum_{n \mid 2^4 \cdot 3^5} 2^{\omega(n)} \sum_{e \mid n} \abs{\mu\parent{n/e}} e^{3/\deg B m}}
+ =& 0\parent{X^{1/(2 \deg B m)} \log X},
```

Submitted Thesis p. 244

```
-... asymptotics ...
+... asymptotics ...

-... of \Cref{Theorem: asymptotic for NQ(X)} for  $m = 10$  ...
+... of \Cref{Theorem: asymptotic for  $\text{twN}(X)$ } for  $m = 10$  ...

-... obtain are the the same ...
+... obtain are the same ...

- We record the order of growth for these modular curves, though not the full asymptotics, in Table \ref{table:modularcurves}.
```

Submitted Thesis p. 245

```
-... curfves ...
+... curves ...

- \twistNad  $m(X)$ , \twconstad  $m X^{\{1/\deg B\ m\}} + O(\text{parent}\{X^{\{1/2\ \deg B\ m\}} \log X\}$ ,
+ \twistNad  $m(X) = \text{twconstad } m X^{\{1/\deg B\ m\}} + O(\text{parent}\{X^{\{1/(2\ \deg B\ m)\}} \log X\}$ ,

-... = 9, 18$.
+... \in \set{6, 9, 18}$.

-... 6, 8, 9, 10, 12, 16, 18}$. ...
+... 6, 8, 9, 12, 16, 18}$. ...

-...  $1/2\ \deg B\ m\}$ .
+...  $1/(2\ \deg B\ m)\}$ .
```

Submitted Thesis p. 246

```
- \mu_{\twistLad  $m\}(\sigma) < 13/84$ 
+ \mu_{\twistLad  $m\}(\sigma) \leq \max\{0, 2-2\deg B\ m\, \sigma\}$ 

- for  $\sigma > 1/2\deg B\ m\}$ .
+ for  $\sigma > 1/(2\ \deg B\ m)\}$ . The same bound holds for  $\text{twistLeq } m(s)\}$ .

-... \set{6, 8, 9, 12, 16, 18}\{m in 4, 6, 8, 9, 12, 16, 18\}\label{Section: ...
+... \set{6, 8, 9, 12, 16, 18}\{m in 6, 8, 9, 12, 16, 18\}\label{Section: ...

-... quadatric ...
+... quadratic ...
```

Submitted Thesis p. 249

```
- Let  $m \in \set{6, 8, 9}\}$ . The sequences
+ Let  $m \in \set{6, 8, 9}\}$  and let  $0 < \delta < 7/97\}$ . The sequences

- are admissible \textup{(\Cref{Definition: admissible sequences})} with parameters  $(1/6, 1/12, 13/42)\}$ .
+ are admissible \textup{(\Cref{Definition: admissible sequences})} with parameters  $(\text{parent}\{1/6, \delta, 97\delta/7\})\}$ .

- The proof is structurally identical to the one given for \Cref{Lemma: Delta NQ(n)} is admissible for  $m = 7\}$ ; however, we must run
  through the argument given there twice, once for  $\text{LQeq } m(s)\}$  and once for  $\text{LQad } m(s)\}$ .
+ The proof is structurally identical to the corrected proof of \Cref{Lemma: Delta NQ(n)} is admissible for  $m = 7\}$ ; however, we
  must run through the argument twice, once for  $\text{LQeq } m(s)\}$  and once for  $\text{LQad } m(s)\}$ .
```

Submitted Thesis p. 250

```
-... \consteqprm  $m X^{\{1/6\}} + O(X^{\{1/8 + \epsilon\}})$ 
+... \consteqprm  $m X^{\{1/6\}} + O(X^{\{38/291 + \epsilon\}})$ 

-... \constadprm  $m X^{\{1/6\}} + O(X^{\{1/8 + \epsilon\}})$ 
+... \constadprm  $m X^{\{1/6\}} + O(X^{\{38/291 + \epsilon\}})$ 

- By \Cref{Lemma: Delta NQ(n)} is admissible for  $5 < m \leq 9\}$ , both  $\text{parent}\{\text{hQeq } m(n)\}_{n \geq 1}\}$  and  $\text{parent}\{\text{hQad } m(n)\}_{n \geq 1}\}$  are admissible with parameters  $(\text{parent}\{1/6, 1/12, 13/42\})\}$ . We now apply \Cref{Theorem: Landau's Tauberian theorem} to the Dirichlet series  $\text{LQeq } m(s)\}$  and  $\text{LQad } m(s)\}$ , and our claim follows.
+ Fix  $\epsilon > 0\}$ . By \Cref{Lemma: Delta NQ(n)} is admissible for  $5 < m \leq 9\}$ , for every  $0 < \delta < 7/97\}$  both  $\text{parent}\{\text{hQeq } m(n)\}_{n \geq 1}\}$  and  $\text{parent}\{\text{hQad } m(n)\}_{n \geq 1}\}$  are admissible with parameters  $(\text{parent}\{1/6, \delta, 97\delta/7\})\}$ . Since  $97\delta/7 < 1\}$ , applying \Cref{Theorem: Landau's Tauberian theorem} and then taking  $\delta$  sufficiently close to  $7/97\}$  gives the error  $O(X^{\{38/291 + \epsilon\}})\}$  in both cases.
```

Submitted Thesis p. 251

```
-... m X^{1/6} \log X + O(X^{1/8} ...
+... m X^{1/6} + O(X^{1/8} ...

-      By \Cref{Lemma: Delta NQ(n)} is admissible for  $m > 9$ , the sequence  $\left(\frac{1}{n} \sum_{m(n) \leq x} \frac{1}{m(n)}\right)_{n \geq 1}$  is admissible with
parameters  $\left(\frac{1}{6}, \frac{1}{12}, \frac{13}{84}\right)$ . We now apply \Cref{Theorem: Landau's Tauberian theorem} to the Dirichlet series  $\sum_{m(n) \leq x} \frac{1}{m(n)}$ , and our claim follows.
+      By \Cref{Lemma: Delta NQ(n)} is admissible for  $m > 9$ , both  $\left(\frac{1}{n} \sum_{m(n) \leq x} \frac{1}{m(n)}\right)_{n \geq 1}$  and  $\left(\frac{1}{n} \sum_{m(n) \leq x} \frac{1}{m(n)^2}\right)_{n \geq 1}$ 
are admissible with parameters  $\left(\frac{1}{6}, \frac{1}{12}, \frac{13}{84}\right)$ . We now apply \Cref{Theorem: Landau's Tauberian theorem} to the
Dirichlet series  $\sum_{m(n) \leq x} \frac{1}{m(n)^2}$  and  $\sum_{m(n) \leq x} \frac{1}{m(n)}$ , and our claim follows.

-... \set{6, 8, 9, 12, 16, 18}\{m in 4, 6, 8, 9, 12, 16, 18\}
+... \set{6, 8, 9, 12, 16, 18}\{m in 6, 8, 9, 12, 16, 18\}

-\calR m (X) = \calR m (1) X^{1/2 \degB m};
+\calR m (X) = \calR m (1) X^{1/(2 \degB m)};
```

Submitted Thesis p. 252

```
-... X^{1/2 \degB m},
+... X^{1/(2 \degB m)},

-... well as how approximately how ...
+... well as approximately how ...
```

Submitted Thesis p. 259

```
-... $65\,68\,1836\,724$ ...
+... $65\,681\,836\,724$ ...
```

Submitted Thesis p. 260

```
-... error, is estimated based on the sampling recorded in Table \ref{table:Rm}.
+... error from the rejection sampling recorded in Table \ref{table:Rm}, is estimated there as well.
```

Submitted Thesis p. 261

```
-... an error term, ...
+... its propagated standard error, ...

-... Error on  $\mathcal{R} m$  ...
+... Standard error ...

-$16$ & $1$ & $4/3$ ...
+$16$ & $2$ & $4/3$ ...

-... $18$ & $2$ & $(1 ...
+... $18$ & $1$ & $(1 ...

-... Error on  $\text{twconsteq } m$  ...
+... Standard error ...

-...  $\text{twconsteq } m$ , its error, and ...
+...  $\text{twconsteq } m$ , its standard error, and ...

-... Error on  $\text{twconstad } m$  ...
+... Standard error ...
```

Submitted Thesis p. 262

```
-...  $\text{twconstad } m$ , its error, and ...
+...  $\text{twconstad } m$ , its standard error, and ...

-$m$ &  $\text{consteq } m$  & Error on  $\text{consteq } m$  &  $\text{constad } m$  & Error on  $\text{constad } m$  \\
+$m$ &  $\text{consteq } m$  & Standard error &  $\text{constad } m$  & Standard error \\
```

Submitted Thesis p. 263

```

-... 3 \zeta(2)$ ...
+... (3 \zeta(2))$ ...

-... / 3 \zeta(2)$, whence ...
+... / (3 \zeta(2))$, whence ...

-$6$ & $0.153\,033\,271$ & $4.8 \cdot 10^{-7}$ & $0.153\,033\,271$ & $4.8 \cdot 10^{-7}$ \\
-$8$ & $0.118\,663\,783$ & $4.4 \cdot 10^{-7}$ & $0.059\,331\,892$ & $2.2 \cdot 10^{-7}$ \\
-$9$ & $0.094\,668\,211$ & $1.7 \cdot 10^{-7}$ & $0.094\,668\,211$ & $1.7 \cdot 10^{-7}$ \\
-\end{tabular}}{The constants $\consteq$ m$ and $\constad$ m$ and their error for $m$ in $\set{6, 8, 9}$}
+$6$ & $0.153\,033\,271$ & $6.1 \cdot 10^{-7}$ & $0.153\,033\,271$ & $6.1 \cdot 10^{-7}$ \\
+$8$ & $0.118\,663\,783$ & $5.5 \cdot 10^{-7}$ & $0.059\,331\,892$ & $2.9 \cdot 10^{-7}$ \\
+$9$ & $0.094\,668\,211$ & $2.1 \cdot 10^{-7}$ & $0.094\,668\,211$ & $2.1 \cdot 10^{-7}$ \\
+\end{tabular}}{The constants $\consteq$ m$ and $\constad$ m$ and their standard errors for $m$ in $\set{6, 8, 9}$}

-&\abs{\int_X^{\infty} \parent{\twistNad 7(u) - \twconst 7 \floor{u^{1/6}}} u^{-7/6} \, \mathrm{d}u} \\
&\abs{\int_X^{\infty} \parent{\twistNad m(u) - \twconst m \floor{u^{1/6}}} u^{-7/6} \, \mathrm{d}u} \\

-We do not know the exact value for $M$, but empirically, we find that for $1 \leq u \leq 10^{42}$, we have
+We do not know the exact value for $M$, but empirically, we find that throughout the range $10^{19} \leq u \leq 10^{42}$, we have

```

Submitted Thesis p. 264

```

-If we assume these bounds continue to hold for larger $u$, and combine this ... \ref{table:ell}, and the estimates for $\consteqpm$
m$ and $\constadpm$ m$ given in Table \ref{table:constm}.
+If we assume these bounds continue to hold for larger $u$, and combine the resulting truncation error with the standard errors in
our approximations for $\twconsteq$ m$ and $\twconstad$ m$, we obtain the uncertainties in Table \ref{table:ell} and Table \ref{
table:constm}.

-$m$ & $\widetilde{ell}_{m, 0}$ & Error on $\widetilde{ell}_{m, 0}$ & $\ell_{m, 0}$ & Error on $\ell_{m, 0}$ \\
+$m$ & $\widetilde{ell}_{m, 0}$ & Uncertainty & $\ell_{m, 0}$ & Uncertainty \\

-$6$ & $-0.572\,182\,786$ & $1.13 \cdot 10^{-1}$ & $-0.636\,153\,135$ & $1.15 \cdot 10^{-1}$ \\ ... $\widetilde{ell}_{m, 0}$ and $\ell_{m, 0}$ and their error for $m$ in $\set{6, 8, 9}$}
+$6$ & $-0.572\,182\,786$ & $1.21 \cdot 10^{-2}$ & $-0.677\,298\,843$ & $1.21 \cdot 10^{-2}$ \\
+$8$ & $-0.742\,599\,375$ & $9.33 \cdot 10^{-3}$ & $-0.387\,595\,153$ & $4.65 \cdot 10^{-3}$ \\
+$9$ & $-0.231\,206\,039$ & $9.37 \cdot 10^{-3}$ & $-0.231\,206\,039$ & $9.37 \cdot 10^{-3}$ \\
+\end{tabular}}{The constants $\widetilde{ell}_{m, 0}$ and $\ell_{m, 0}$ and their numerical uncertainties for $m$ in $\set{6, 8, 9}$}
}

-$m$ & $\consteqpm$ m$ & Error on $\consteqpm$ m$ & $\constadpm$ m$ & Error on $\constadpm$ m$ \\
+$m$ & $\consteqpm$ m$ & Uncertainty & $\constadpm$ m$ & Uncertainty \\

-$6$ & $-0.037\,215\,321$ & $1.78 \cdot 10^{-1}$ & $-0.114\,993\,938$ & $1.78 \cdot 10^{-1}$ \\ ... $\consteqpm$ m$ and $\constadpm$
m$ and their error for $m$ in $\set{6, 8, 9}$}
+$6$ & $-0.037\,215\,321$ & $1.47 \cdot 10^{-2}$ & $-0.165\,021\,121$ & $1.47 \cdot 10^{-2}$ \\
+$8$ & $-0.392\,302\,971$ & $1.14 \cdot 10^{-2}$ & $-0.215\,964\,396$ & $5.65 \cdot 10^{-3}$ \\
+$9$ & $0.126\,227\,997$ & $1.14 \cdot 10^{-2}$ & $0.126\,227\,997$ & $1.14 \cdot 10^{-2}$ \\
+\end{tabular}}{The constants $\consteqpm$ m$ and $\constadpm$ m$ and their numerical uncertainties for $m$ in $\set{6, 8, 9}$}

```

Submitted Thesis p. 265

```

-\frac{\twistNeq m(X)}{X^{1/6}} ...
+\frac{NNeq m(X)}{X^{1/6}} ...

-\frac{\twistNad m(X)}{X^{1/6}} ...
+\frac{NQad m(X)}{X^{1/6}} ...

-... $\frac{\twistNeq}{...}$
+... $\frac{NNeq}{...}$

-... $\frac{\twistNad}{...}$
+... $\frac{NQad}{...}$

-... $146\,844\,192$ & $-0.115\,210\,235$ ...
+... $146\,343\,642$ & $-0.165\,265\,235$ ...

-... $55\,418\,512$ & $-0.196\,051\,423$ ...
+... $55\,219\,676$ & $-0.215\,935\,023$ ...

```

Submitted Thesis p. 266

```

-... 2 \twistNeq ...
+... 2 \twistLeq ...

-... 2 \twistNad {m}(1/6)/\zeta(2),
+... 2 \twistLad {m}(1/6)/\zeta(2),

```

```

-... 0.212\,842\,775\,719\,189\,16, ...
+... 0.212\,842\,775\,719\,150\,50, ...

-... 0.106\,421\,387\,859\,584\,87, ...
+... 0.106\,421\,387\,859\,575\,25, ...

-... 0.269\,169\,745\,679\,629\,80, ...
+... 0.269\,169\,745\,679\,529\,87, ...

-... 0.134\,584\,872\,839\,728\,06, ...
+... 0.134\,584\,872\,839\,764\,93, ...

-... 0.107\,025\,809\,031\,522\,12.
+... 0.107\,025\,809\,032\,944\,72.

```

Submitted Thesis p. 267

```

-If these bounds continue to hold for larger  $X$ , then
+If these bounds continue to hold for larger  $X$ , then partial summation, using the endpoint values from Table \ref{table:enumerating}, gives

-\sum_{n > 10^{84}} \frac{\#\{12\}(n)}{n^{1/6}} \approx \int_{10^{84}}^{\infty} x^{-1/6} d \#\{12\}(x) < 1.090\,472 X^{1/12} ...
  10^{126}} \frac{\#\{18\}(n)}{n^{1/6}} \approx \int_{10^{126}}^{\infty} x^{-1/6} d \#\{18\}(x) < 5.035\,48 \cdot 10^{-15}.
  \\\
+\sum_{n > 10^{84}} \frac{\#\{12\}(n)}{n^{1/6}} < 1.133\,046\,8 \cdot 10^{-7}, \\\
+\sum_{n > 10^{84}} \frac{\#\{12\}(n)}{n^{1/6}} < 5.665\,234 \cdot 10^{-8}, \\\
+\sum_{n > 10^{84}} \frac{\#\{16\}(n)}{n^{1/6}} < 9.118\,462 \cdot 10^{-8}, \\\
+\sum_{n > 10^{84}} \frac{\#\{16\}(n)}{n^{1/6}} < 4.559\,231 \cdot 10^{-8}, \text{ and } \\\
+\sum_{n > 10^{126}} \frac{\#\{18\}(n)}{n^{1/6}} < 5.375\,8 \cdot 10^{-15}. \\\

-... estimates ...
+... estimates ...

- $m$  &  $\text{consteq } m$  & Error on  $\text{consteq } m$  &  $\text{constad } m$  & Error on  $\text{constad } m$  \\\
+ $m$  &  $\text{consteq } m$  & Error bound &  $\text{constad } m$  & Error bound \\\

- $12$  &  $0.258\,785\,783$  &  $1.4 \cdot 10^{-7}$  &  $0.129\,392\,891$  &  $6.7 \cdot 10^{-8}$  \\\ ...  $\text{consteq } m$  and  $\text{constad } m$  and
  their error for  $m \in \{12, 16, 18\}$ 
+ $12$  &  $0.258\,785\,784$  &  $1.4 \cdot 10^{-7}$  &  $0.129\,392\,892$  &  $6.9 \cdot 10^{-8}$  \\\
+ $16$  &  $0.327\,271\,167$  &  $1.2 \cdot 10^{-7}$  &  $0.163\,635\,583$  &  $5.6 \cdot 10^{-8}$  \\\
+ $18$  &  $0.130\,127\,779\,817\,961\,2$  &  $6.6 \cdot 10^{-15}$  &  $0.130\,127\,779\,817\,961\,2$  &  $6.6 \cdot 10^{-15}$ 
+\end{tabular}}{The constants  $\text{consteq } m$  and  $\text{constad } m$  and their conditional error bounds for  $m \in \{12, 16, 18\}$ }

```

Submitted Thesis p. 268

```

-In each of these cases, the compactified moduli space  $X_0(m)$  is of genus  $g > 0$ , so by Faltings's theorem (\Cref{Theorem:
  Faltings Theorem}), each curve has a finite number of rational points.
+In each of these cases, the compactified moduli space  $X_0(m)$  has genus  $g > 0$  and only finitely many rational points.

```

Submitted Thesis p. 269

```

-... curves admitting cyclic  $m$ -isogeny, ...
+... curves admitting a cyclic  $m$ -isogeny, ...

-... 37, 43, 163}.
+... 37, 43, 67, 163}.

```

Submitted Thesis p. 270

```

-The following table is extracted from \cite[pag 1]{Mazur2}.
+The twist classes and their  $j$ -invariants are classical; see \cite[pag 1]{Mazur2} and the references there. We use twist-minimal
  short Weierstrass representatives with  $B > 0$ .

-...  $(-1\,8075, \dots$ 
+...  $(-18\,075, \dots$ 

- $27$  &  $(-270, -1\,708)$  &  $78\,766\,128$  &  $-12288000$  \\\
+ $27$  &  $(-120, 506)$  &  $6\,912\,972$  &  $-12288000$  \\\

-...  $730\,6267\,500$  ...
+...  $7\,306\,267\,500$  ...

-...  $6\,313\,1603\,150$  ...
+...  $63\,131\,603\,150$  ...

- $43$  &  $(-3\,440, 42)$  &  $162\,830\,336\,000$  &  $-884736000$  \\\

```

+\$43\$ & \$(-3\,440, 77\,658)\$ & \$162\,830\,654\,028\$ & \$-884736000\$ \\\

Submitted Thesis p. 271

-By the Mazur's theorem ...
+By Mazur's theorem ...

... Table \ref{table:nonzerogenus} it is an ...
+... Table \ref{table:nonzerogenus} is an ...

... curves admitting cyclic m -isogeny ...
+... curves admitting a cyclic m -isogeny ...

... 2, 3, 4, 5, 6, 8, 9, 10, 12, ...
+... 2, 3, 4, 5, 6, 7, 8, 9, 10, 12, ...

Submitted Thesis p. 272

... 78\,766\,128), ...
+... 6\,912\,972), ...

... 730\,6267\,500) ...
+... 7\,306\,267\,500) ...

... 162\,830\,336\,000), ...
+... 162\,830\,654\,028), ...

Submitted Thesis p. 273

... additional ...
+... additional ...

... 37, 43, 163},
+... 37, 43, 67, 163},

Submitted Thesis p. 275

-\Nqeq {11} (X) = & \kfree 2\parent{\parent{X/77\,480\,172}^{1/6}} + \kfree 2\parent{\parent{X/2\,923\,690\,572}^{1/6}} \\\

-&+ \kfree 2\parent{\parent{X/323\,181\,166\,572}^{1/6}}, \\\

+\Nqeq {11} (X) = & 2 \kfree 2\parent{\parent{X/77\,480\,172}^{1/6}} + 2 \kfree 2\parent{\parent{X/2\,923\,690\,572}^{1/6}} \\\

+&+ 2 \kfree 2\parent{\parent{X/323\,181\,166\,572}^{1/6}}, \\\

- \Nqeq {14} (X) = & \kfree 2\parent{\parent{X/259\,308}^{1/6}} + \kfree 2\parent{\parent{X/842\,579\,500}^{1/6}}, \\\

+\Nqeq {14} (X) = & 2 \kfree 2\parent{\parent{X/259\,308}^{1/6}} + 2 \kfree 2\parent{\parent{X/842\,579\,500}^{1/6}}, \\\

- \Nqeq {15}(X) = & \kfree 2\parent{\parent{X/234\,967\,500}^{1/6}} + \kfree 2\parent{\parent{X/478\,550\,700}^{1/6}} \\\

-&+ \kfree 2\parent{\parent{X/126\,818\,068\,500}^{1/6}} + \kfree 2\parent{\parent{X/23\,621\,749\,807\,500}^{1/6}}, \\\

+\Nqeq {15}(X) = & 2 \kfree 2\parent{\parent{X/234\,967\,500}^{1/6}} + 2 \kfree 2\parent{\parent{X/478\,550\,700}^{1/6}} \\\

+&+ 2 \kfree 2\parent{\parent{X/126\,818\,068\,500}^{1/6}} + 2 \kfree 2\parent{\parent{X/23\,621\,749\,807\,500}^{1/6}}, \\\

- \Nqeq {17} (X) = & \kfree 2\parent{\parent{X/4\,325\,629\,743\,607\,500}^{1/6}} + \kfree 2\parent{\parent{X/335\,734\,876\,712\,407\,500}^{1/6}}, \\\

+\Nqeq {17} (X) = & 2 \kfree 2\parent{\parent{X/4\,325\,629\,743\,607\,500}^{1/6}} + 2 \kfree 2\parent{\parent{X/335\,734\,876\,712\,407\,500}^{1/6}}, \\\

... (X) = & \kfree 2\parent{\parent{X/14\,074\,668}^{1/6}}, ...

+... (X) = & 2 \kfree 2\parent{\parent{X/14\,074\,668}^{1/6}}, ...

- \Nqeq {21}(X) = & \kfree 2\parent{\parent{X/364\,500}^{1/6}} + \kfree 2\parent{\parent{X/1\,853\,388}^{1/6}} \\\

-&+ \kfree 2\parent{\parent{X/57\,395\,607\,372}^{1/6}} + \kfree 2\parent{\parent{X/20\,478\,321\,699\,372}^{1/6}}, \\\

+\Nqeq {21}(X) = & 2 \kfree 2\parent{\parent{X/364\,500}^{1/6}} + 2 \kfree 2\parent{\parent{X/1\,853\,388}^{1/6}} \\\

+&+ 2 \kfree 2\parent{\parent{X/57\,395\,607\,372}^{1/6}} + 2 \kfree 2\parent{\parent{X/20\,478\,321\,699\,372}^{1/6}}, \\\

- \Nqeq {27} (X) = & \kfree 2\parent{\parent{X/78\,766\,128}^{1/6}}, \\\

+\Nqeq {27} (X) = & 2 \kfree 2\parent{\parent{X/6\,912\,972}^{1/6}}, \\\

- \Nqeq {37} (X) = & \kfree 2\parent{\parent{X/730\,6267\,500}^{1/6}} + \kfree 2\parent{\parent{X/107\,611\,181\,539\,805\,427\,907\,500}^{1/6}}, \\\

+\Nqeq {37} (X) = & 2 \kfree 2\parent{\parent{X/7\,306\,267\,500}^{1/6}} + 2 \kfree 2\parent{\parent{X/107\,611\,181\,539\,805\,427\,907\,500}^{1/6}}, \\\

- \Nqeq {43} (X) = & \kfree 2\parent{\parent{X/162\,830\,336\,000}^{1/6}}, \\\

+\Nqeq {43} (X) = & 2 \kfree 2\parent{\parent{X/162\,830\,654\,028}^{1/6}}, \\\

... (X) = & \kfree 2\parent{\parent{X/102\,480\,782\,771\,052}^{1/6}}, ...

+... (X) = & 2 \kfree 2\parent{\parent{X/102\,480\,782\,771\,052}^{1/6}}, ...

-... (X) =& \kfree 2\parent{\parent{X/2\,631\,905\,352\,272\,628\,650\,988}~{1/6}}.
+... (X) =& 2 \kfree 2\parent{\parent{X/2\,631\,905\,352\,272\,628\,650\,988}~{1/6}}.

-=& \kfree 2 ...
+=& 2 \kfree 2 ...

Submitted Thesis p. 276

-... the present ...
+... the presence ...

Submitted Thesis p. 277

-\$11\$ & \$0.05285852537804229\ldots\$ \\\ \$14\$ & \$0.09590984282353528\ldots\$ \\\ \$15\$ & \$0.05837531634681239\ldots\$ \\\ \$17\$ & ... \$0
.013888883070281625\ldots\$ \\\ \$43\$ & \$0.00822676234970696\ldots\$ \\\ \$67\$ & \$0.002810246610438085\ldots\$ \\\ \$163\$ & \$0
.00016360872509265466\ldots\$ \\\
+\$11\$ & \$0.10571705075608457\ldots\$ \\\
+\$14\$ & \$0.19181968564707050\ldots\$ \\\
+\$15\$ & \$0.11675063269362475\ldots\$ \\\
+\$17\$ & \$0.0044705452369290256\ldots\$ \\\
+\$19\$ & \$0.07824834140601367\ldots\$ \\\
+\$21\$ & \$0.28048805576004344\ldots\$ \\\
+\$27\$ & \$0.08809218206211060\ldots\$ \\\
+\$37\$ & \$0.027777766140563242\ldots\$ \\\
+\$43\$ & \$0.01645351934345495\ldots\$ \\\
+\$67\$ & \$0.005620493220876168\ldots\$ \\\
+\$163\$ & \$0.00032721745018530919\ldots\$ \\\